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HASHEMI AMER » - BOUDOUAOU



Final Graduation Project To obtain the degree of: Senior Technician
(BTS) in Arts and Graphic Industry

Option: Study and Production of Printed Products

THEME:



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DEDICATION

To Allah, the Almighty, I thank Allah for giving me the strength, health, and patience to complete this work. Thank You for Your blessings and for helping me succeed in my studies. To my dear Parents, to my beloved Mother, thank you for your love, your prayers, and everything you have done for me. To my beloved Father, thank you for your support, your sacrifices, and for always believing in me. I dedicate this work to both of you with all my love and respect. To my beloved Family, to my dear brothers and sisters, thank you for your encouragement and for always being by my side. To all my family members, thank you for your love and support. You are the most important people in my life. To my Friends and Teachers, to my friends, thank you for the wonderful moments, your help, and your support during my studies. To my teachers at the National Specialized Institute (Hashemi Amer – Boudouaou), thank you for sharing your knowledge and teaching me the skills of the printing industry. Finally, I dedicate this work to everyone who helped me and supported me throughout my studies.

Contents

Dedication	2
LIST OF FIGURES	7
LIST OF TABLES	11
General Introduction	12
.....	14
PART I :	14
BACKGROUND OF THE REPORT	14
.....	15
CHAPTER N°1:	15
PRESENTATION OF THE COMPANY (SIA)	15
Introduction	16
1. Who is SIA?	16
2. History of the Company	17
4. Company Organization	19
5. Services	21
6. Equipment and Tools	23
.7 Main Clients of the Company	30
.8 Main Products of the Company	30
Chapter Conclusion	30
CHAPTER N°2:	31
PROFESSIONAL IMMERSION	31
Introduction	32
1. Departments Visited During the Internship	32
2. Tasks and Missions	39
.....	43
PART II :	43

PROFESSIONAL PROJECT	43
CHAPTER N°3 :.....	44
PROJECT PREPARATION	44
Introduction.....	45
1. Project Presentation	45
2. Project Objectives.....	46
3. Production Workflow	46
4. Technical Specifications.....	48
5. Manufacturing Cost Calculation.....	49
PRODUCTION COST CALCULATION	49
5. Manufacturing File	57
Conclusion	59
CHAPTER N°4 :.....	59
PROJECT IMPLEMENTATION	59
.....	60
SECTION 01 :.....	60
PREPRESS	60
Introduction.....	61
1. Prepress Department.....	61
2. Prepress Workflow	61
.4 Verification and Processing of the PDF File.....	62
.5 Structural Design of the Packaging.....	64
5.3 Adjustment of the Hanging Hole and Cutting Lines	66
6. Elaboration of the B.A.T	69
7. Printing and Approval of the B.A.T	74
.8 Plate Making	75

Chapter Conclusion.....	86
.....	87
SECTION 02 :.....	87
PRESS.....	87
Chapter Introduction	88
1. Heidelberg Speedmaster XL 106.....	88
2. The materials used	89
3. The printing steps of the packaging.....	89
4. Printing Make-Ready.....	93
5. Printing Part:.....	102
Chapter Conclusion.....	112
SECTION 03 :.....	113
POST-PRESS.....	113
Introduction.....	114
.1 Die-cutting.....	114
Value	115
Folding and gluing	127
.2 Folding and gluing machine	127
Value	128
Chapter Conclusion.....	137
General Conclusion.....	138
General Conclusion.....	139
ACKNOWLEDGEMENTS.....	140
REFERENCES	141
ANNEXES.....	142

LIST OF ABBREVIATIONS

- ✚ CMYK: Cyan, Magenta, Yellow, Black.
- ✚ RGB: Red, Green, Blue.
- ✚ RIP: Raster Image Processor.
- ✚ DTP: Desktop Publishing.
- ✚ DPI: Dots Per Inch.
- ✚ LPI: Lines Per Inch.
- ✚ ICC: International Color Consortium.
- ✚ ISO: International Organization for Standardization.
- ✚ AM: Amplitude Modulation.
- ✚ PDF: Portable Document Format.
- ✚ CTP: Computer-to-Plate.
- ✚ CTF: Computer-to-Film.
- ✚ BAT: Print Approval Proof.
- ✚ BAR: Press Approval.
- ✚ HT: Excluding Tax.
- ✚ VAT: Value Added Tax.
- ✚ TTC: Including Tax.
- ✚ DA: Algerian Dinar.
- ✚ FIG.: Figure.
- ✚ FSC: Forest Stewardship Council.

LIST OF FIGURES

Figure 1. Main entrance of Société d'Impression d'Alger (SIA)	17
Figure 2 . Location of Société d'Impression d'Alger (SIA)	19
Figure 3 . illustrates the general organizational structure of SIA, showing the distribution of responsibilities from the top management to the operational departments and regional units.	20
Figure 4 . Equipment Layout of the Rotary Workshop	26
Figure 5. Layout of the Equipment in the Labeur Workshop	27
Figure 6. Layout of the Equipment in the Packaging Workshop.....	28
Figure 7. Layout of the Equipment in the Pre-press Workshop	29
Figure 8. Main Clients of the Company.....	30
Figure 9. Main Products of the Company	30
Figure 10. Pre-press Workshop.....	32
Figure 11. Checking and Preparing Customer Files	33
Figure 12. Computer-to-Plate (CTP) System.....	33
Figure 13. Control Panel of the Heidelberg XL106.....	34
Figure 14. Heidelberg XL106	35
Figure 15. Die-cutting Process.....	36
Figure 16. MK Easymatrix 106 CS Die-cutting Machine	37
Figure 17. MK Diana Easy 115 Folding and Gluing Machine	38
Figure 18. Finished Packaging Boxes.....	38
Figure 19. File Verification in Prinect Package Designer	39
Figure 20. Screenshot of the CTP software	40
Figure 21. Printing Plates.....	40
Figure 22. Finished School Communication Book.....	41
Figure 23. Magma Printer's Cable Packaging Box	45
Figure 24. Production Workflow of the Magma Packaging Box	47
Figure 25. Parallel Production Workflow of the Magma Printer's Cable Packaging Box.....	57
Figure 26. Manufacturing File Used for the Production of the Magma Printer's Cable Packaging Box	58
Figure 27. Prepress workflow of the Magma Printer's Cable Packaging Box.....	61
Figure 28. Final PDF artwork received from the customer	62

Figure 29. Opening the PDF file in Adobe Illustrator and checking the compatibility with the die-cut template.....	63
Figure 30. PDF/X-4 export according to SIA production standards	64
Figure 31. Selection of the Packaging Template (EVF11065)	65
Figure 32. Adjustment of the Packaging Dimensions	66
Figure 33. Adjustment of the Hanging Hole and Cutting Lines	66
Figure 34. Positioning the Artwork on the Packaging Layout.....	67
Figure 35. Three-Dimensional Visualization of the Packaging.....	68
Figure 36. Saving and Exporting the Packaging Project	68
Figure 37. Creating a new production job and preparing the workspace in Prinect Signa Station Source: Taken by the author.	69
Figure 38. Importing the die-cut file into Prinect Signa Station and loading it into the project	70
Figure 39. Setting the first packaging layout in Prinect Signa Station	70
Figure 40. Creating the printing layout for the packaging.....	71
Figure 41. Importing the final artwork into the packaging layout	72
Figure 42. Adjusting the artwork position on the die-cut	73
Figure 43. Final verification of the complete printing layout	73
Figure 44. Saving and exporting the final production file	74
Figure 45. HP DesignJet T520 printer used to print the B.A.T for packaging verification....	75
Figure 46. APOGEE Prepress main interface	75
Figure 47. Magma packaging project inside APOGEE	76
Figure 48. Color separation of the Magma packaging.....	77
Figure 49. Starting the processing by selecting Continue Processing	77
Figure 50. Green blinking output icon during plate processing.....	78
Figure 51. CTP machine screen showing the plate path inside the machine	79
Figure 52. Plate Engraving in Progress.....	79
Figure 53. Loading the printing plate into the CTP machine	80
Figure 54. Engraved plate moving to the developing unit.....	81
Figure 55. Chemical solutions used in the Avalon N8-80 processor.....	82
Figure 56. Chemical tanks connected to the Avalon N8-80 processor.....	82
Figure 57. Development temperature setting on the Avalon N8-80 processor screen	83
Figure 58. Plate development inside the Avalon N8-80 processor.....	83
Figure 59. Plate leaving the processor and plate storage	84

Figure 60. Plate punching on the Heidelberg machine.	85
Figure 61. Plate punching and plate bending.....	85
Figure 62. Measurement of the printing plate using the Techkon SpectroDens.....	86
Figure 63. Different units of the Heidelberg Speedmaster XL 106.	88
Figure 64. Magma production file	90
Figure 65. Cleaning of the plate cylinder before printing.....	91
Figure 66. pH scale used to check the acidity of the dampening solution.....	92
Figure 67. Dampening solutions used for the preparation of the printing press.....	92
Figure 68. Resetting the registers.....	93
Figure 69. Feeder adjustment.....	94
Figure 70. Air cushion blowers.....	94
Figure 71. Sheet separation blowers	95
Figure 72. Separation blades.....	95
Figure 73. Lifting suckers and Transport suckers.....	96
Figure 74. Presser feet.....	96
Figure 75. Delivery	98
Figure 76. Blank run	99
Figure 77. Printing Functions settings	99
Figure 78. Work data and printing speed.....	100
Figure 79. Manual ink profile reading on the light table.	101
Figure 80. Ink profile tab	101
Figure 81. Anti-set-off powder	102
Figure 82. Ink fountain filling.....	102
Figure 83. Cleaner.....	103
Figure 84. Plate mounting on the offset press.....	104
Figure 85. Printing unit control buttons.....	105
Figure 86. Loupe.....	105
Figure 87. Color registration.....	106
Figure 88. Register menu.....	106
Figure 89. Pre-inking settings	108
Figure 90. Varnishing preparation	110
Figure 91. Printed sheet	111
Figure 92. Ink density measurement with X-Rite densitometer	111
Figure 93. MK Easymatrix die-cutting machine.....	114

Figure 94. Die-cutting form	116
Figure 95. Creasing calculation and optimum creasing system.....	117
Figure 96. Guillotine	118
Figure 97. Creasing counterparts	118
Figure 98. Foams	119
Figure 99. Adhesive tape	119
Figure 100. Die-cutting form with channels	120
Figure 101. Makeready sheet	120
Figure 102. Creasing rule protection	121
Figure 103. Braille counterparts and channels.....	121
Figure 104. MK Easymatrix 106 CS control panel – machine settings.....	122
Figure 105. Feeder adjustment.....	123
Figure 106. Feeder head.....	124
Figure 107. Feed table roller adjustment.	124
Figure 108. Delivery rear stop adjustment.....	125
Figure 109. Waste separation machine – 4500 rpm.....	126
Figure 110. Manual waste separation	127
Figure 111. MK Diana Easy folding and gluing machine	128
Figure 112. MK Diana Easy	129
Figure 113. Glue used for the Magma box.	129
Figure 114. First sheet placement and feeder adjustment.....	130
Figure 115. First pre-breaking	131
Figure 116. Second pre-breaking.....	131
Figure 117. Box passage through the transporters.....	132
Figure 118. Gluing system.....	132
Figure 119. Box closing (folding).....	133
Figure 120. Box passage through the transporters.....	134
Figure 121. Box passage through the press.	134
Figure 122. Folding and gluing machine screen settings.....	135
Figure 123. Filling the cartons with the finished boxes.....	135
Figure 124. Cartons ready for delivery.	136
Figure 125. Final Magma product	136

LIST OF TABLES

Table 1. Technical Specifications	48
Table 2. Production schedule of the Magma Printer's Cable Packaging Box.....	57
Table 3. Problems, causes and solutions during printing	112
Table 4. Problems, causes and solutions during die cutting.	127
Table 5. Folding and gluing problems	137

GENERAL INTRODUCTION

The graphics industry is an important sector that includes all the people, equipment, and technologies involved in the production of printed products. The production process starts with the preparation and design of the artwork (Pre-press), continues with the printing stage (Press), and ends with the finishing operations (Post-press). This industry also depends on different materials such as paper, printing plates, inks, proofs, and other consumables that are necessary to produce high-quality printed products.

Packaging is an essential part of the graphics industry. The word "Packaging" refers to the design and production of containers used to protect, transport, and present products. It includes different types of packages such as boxes, labels, bags, bottles, and other packaging materials. Packaging has two main functions. The first is to protect the product from damage, humidity, dust, and other external factors during transportation and storage. The second is to improve the appearance of the product, provide useful information to customers, and strengthen the visual identity of the brand.

This graduation project, entitled "Study and Production of Magma Printer Cable Packaging Box," was carried out at the Société d'Impression d'Alger (SIA), one of the largest printing companies in Algeria. The company is equipped with modern technologies, advanced software, and high-performance printing machines that ensure the production of high-quality packaging.

The main objective of this project is to study and produce the packaging box of the Magma Printer Cable using the offset printing process. The work covers all the production stages, including structural design, pre-press preparation, offset printing, and post-press finishing until the final packaging box is completed.

The main question of this study is: How can a high-quality packaging box be produced using the offset printing process while respecting the required technical and production standards? To answer this question, different technical documents, professional references, and the practical experience of the staff at SIA were used throughout this project.

This thesis is divided into two main parts. The first part presents the company and describes the professional internship carried out at SIA. The second part focuses on the technical study and the production of the Magma Printer Cable Packaging Box. It explains the project preparation, the technical specifications, and the different production stages, including Pre-press, Press, and Post-press operations.

Finally, this work presents the complete production process of the Magma packaging box and highlights the importance of good planning, modern printing technologies, and quality control in producing a professional packaging product.



PART I :
BACKGROUND OF THE REPORT



CHAPTER N°1:
PRESENTATION OF THE COMPANY (SIA)

Introduction

This chapter presents Société d'Impression d'Alger (SIA), the company where the practical training was carried out. It gives a general idea about the company, its history, organization, services, equipment, clients, and main products.

First, the chapter introduces SIA and gives some information about its creation, location, activities, and development. Then, it presents the different directions and departments of the company and explains their main roles. The chapter also describes the main workshops and the equipment used in pre-press, printing, packaging, and newspaper production.

Finally, this chapter presents some of the main clients and products of SIA. This information helps the reader understand the company, its activities, and the different stages and resources involved in its printing and packaging production.

1. Who is SIA?

Société d'Impression d'Alger (SIA) is a public printing company in Algeria. It works as a Public Economic Enterprise (EPE) and a Joint-Stock Company (SPA). The company is well known for newspaper printing and many other printing services.

SIA was created on 11 December 1990 after the reorganization of the former Ech-Chaab company. Its head office is in the Industrial Zone of El Alia, Bab Ezzouar, Algiers. The company has a share capital of 1,500,000,000 DA and more than 300 employees.

Besides its main office in Algiers, SIA has two production units in Ouargla and Bechar. These units help the company provide its services in different parts of Algeria.

SIA prints many products such as newspapers, books, magazines, packaging boxes, and other printed materials. The company uses modern machines and printing technology to produce good-quality products that meet customer needs.

Today, SIA is one of the main printing companies in Algeria. It continues to improve its equipment and production methods to provide better printing and packaging services.



Figure 1. Main entrance of Société d'Impression d'Alger (SIA) .

Source: Internet.

2. History of the Company

Société d'Impression d'Alger (SIA) was created on **11 December 1990** after the restructuring of the former **Ech-Chaab** company. From the beginning, the company worked to improve its production and update its printing equipment.

In **1992**, SIA increased its newspaper printing capacity by installing a new **Harris Heidelberg rotary press**. Later, in **2002**, the company expanded its facilities by acquiring a new site in **Bab Ezzouar** and installing a **Goss Universal rotary press**.

In **2003**, SIA started printing more than newspapers. Its production included school textbooks, books, magazines, and many other printed products. One year later, in **2004**, the company became the national leader in newspaper printing, producing about **44%** of all newspapers printed in Algeria.

To improve print quality, SIA introduced **Computer-to-Plate (CTP)** technology in **2005**. In **2008**, the company opened its first regional production unit in **Ouargla**. In **2009**, it installed a **Goss Community rotary press** to increase its production capacity.

The company continued to modernize its workshops. In **2010**, it installed new equipment for cutting, folding, stitching, and binding. In **2012**, SIA added a modern **five-color Heidelberg offset printing machine with varnish**, which further improved the quality of its printed products.

The company's growth continued in **2013** with the opening of its second regional production unit in **Bechar**. In **2014**, SIA expanded its pre-press department by installing **two new AGFA Computer-to-Plate (CTP) systems**, showing its commitment to modern printing technology and continuous improvement.

3. Company Information

Société d'Impression d'Alger (SIA) is an Algerian printing company specialized in printing and packaging production. The company is located in the industrial zone of El Alia, Bab-Ezzouar, Algiers, Algeria.

The company's contact information is as follows:

Address: Zone industrielle El Alia, Bab-Ezzouar, Alger.

Tel./Fax: +213 (0) 23 839 876

Mobile: +213 (0) 23 839 675

Email: contact@sia.dz

Website: <https://www.sia.dz>

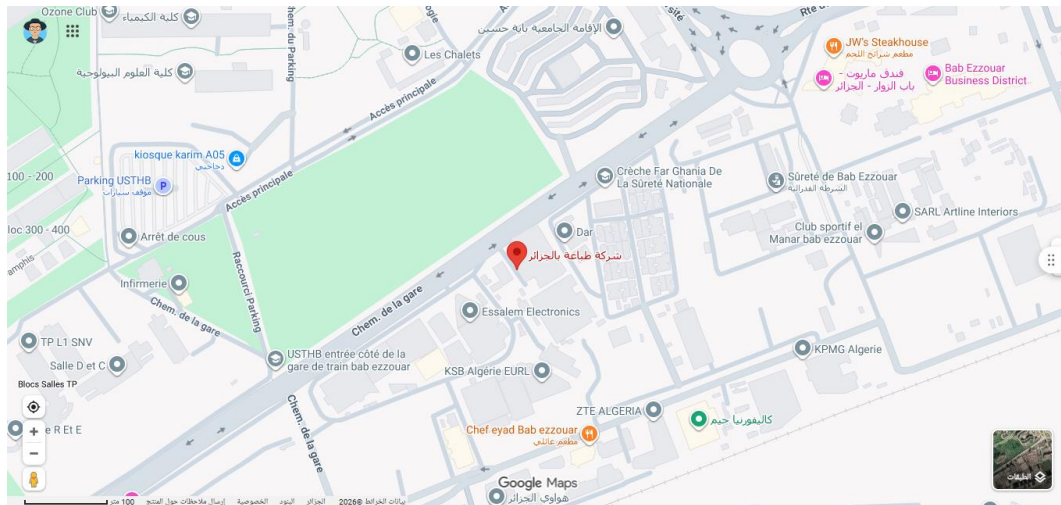


Figure 2 . Location of Société d'Impression d'Alger (SIA)

Source: Internet.

4. Company Organization

The organizational structure of Société d'Impression d'Alger (SIA) shows how the company is managed and how the different departments work together. It defines the responsibilities of each management level and helps organize the company's daily work. The company is headed by the President Director General, who is supported by several administrative and control functions. Under the general management, the company is divided into several main directions, each responsible for a specific area of activity. These directions include different departments and services that work together to support the company's production and management activities. Figure 3 shows the organizational chart of Société d'Impression d'Alger (SIA).

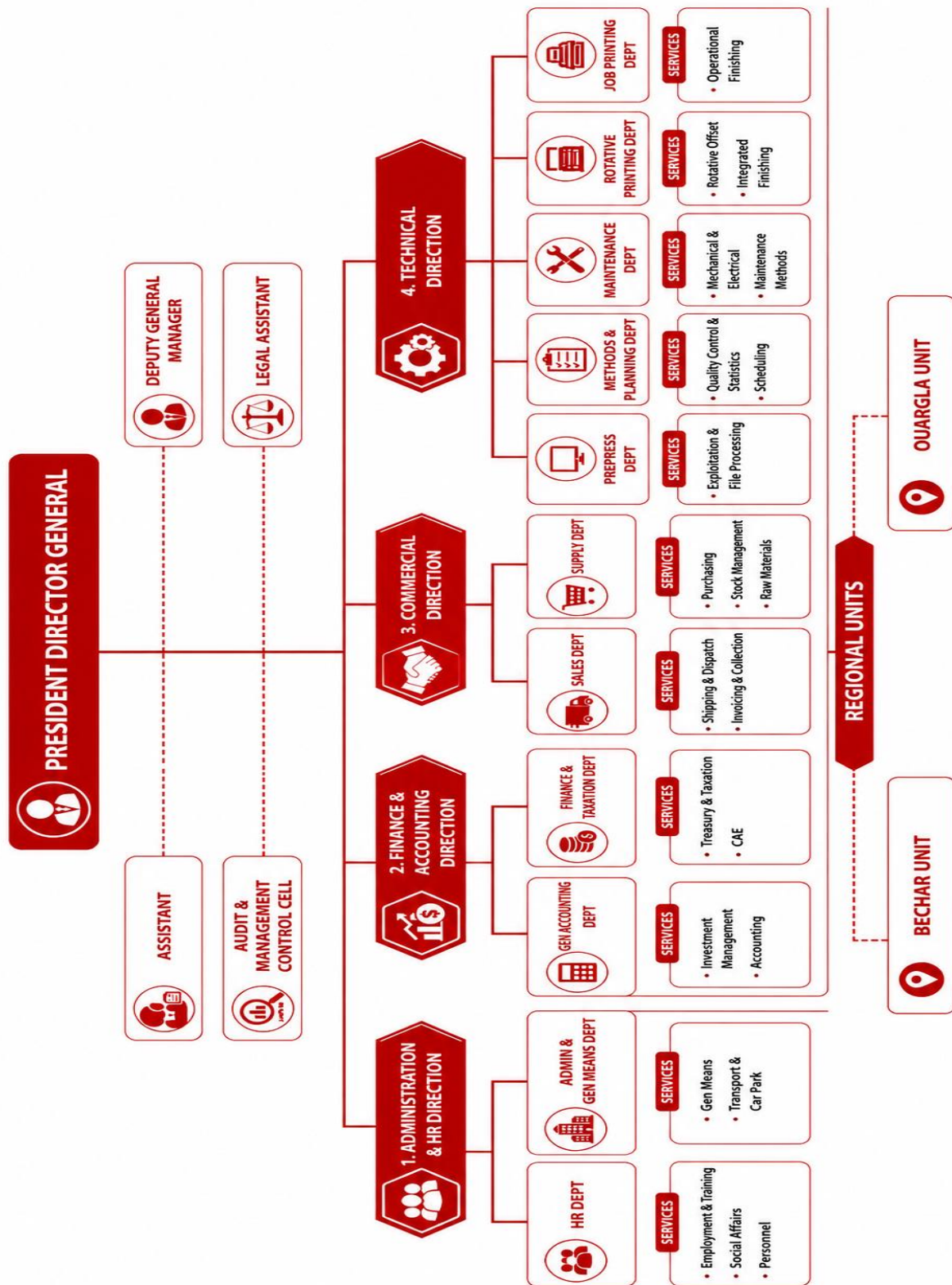


Figure 3 . illustrates the general organizational structure of SIA, showing the distribution of responsibilities from the top management to the operational departments and regional units.

Source: Designed by the author.

5. Services

The work at Société d'Impression d'Alger (SIA) is divided into several main directions. Each direction has a specific role and helps the company manage its daily activities. Together, these directions support the production process and ensure that the company works efficiently.

5.1 Administration and Human Resources Direction

This direction manages the company's employees. It is responsible for recruitment, training, employee careers, salaries, and social services.

Departments:

Human Resources Department

- Employment and Training Service
- Social Affairs Officer
- Personnel Service

Administration and General Means Department

- General Means Service
- Car Park Service

5.2 Finance and Accounting Direction

This direction is responsible for invoice accounting, regular monitoring of customer debts and payment deadlines, management of financial operations, and general, analytical, and budget accounting. These activities allow the company to prepare its operating account and financial statements.

It includes:

General Accounting Department

- Investment Management Officer
- Accounting Officers

Finance and Taxation Department

- Treasury and Taxation Officer
- CAE Analysis Officer

5.3 Commercial Direction

The Commercial Direction is responsible for the company's commercial activities. It works to find new customers, maintain good relationships with existing customers, and increase sales. It also manages the supply of raw materials needed for production.

Departments:

Sales Department

- Shipping Service
- Billing and Collection Service

Supply Department

- Purchasing Service
- Stock Management Service
- Raw Material Preparation Service

5.4 Technical Direction

The Technical Direction is responsible for all production activities. It starts with file preparation before printing and continues through printing, finishing, and maintenance. This direction also controls production planning and product quality to ensure that the work is completed correctly.

Departments:

- Pre-press Department
- Production, Planning, Quality and Statistics Department
- Maintenance Department
- Printing Department
- Labeur Department

6. Equipment and Tools

Société d'Impression d'Alger (SIA) has many modern machines to support large-scale printing production. These machines are installed in different workshops according to their functions.

6.1 Pre-press Workshop

The Pre-press Workshop is responsible for preparing printing plates and digital files before the printing process. It uses modern equipment to ensure high printing quality.

Equipment

- 03 CTP machines for newspapers: AGFA ADVANTAGE N-TR.
- 01 CTP machine for publishing: AGFA ELANTRIX 125 HX.
- 03 MAC computers.
- 02 IMAC computers.
- 01 Digital Proofer: EPSON Stylus Pro 7900.
- 01 Digital Proofer: HP.
- 02 Tables.
- 05 DELL computers.

6.2 Printing Workshop (Labeur)

The Printing Workshop is responsible for printing, folding, cutting, perforating, and binding different printed products.

Equipment

- 01 Offset press with 5 printing units including varnish: HEIDELBERG SM 102.
- 01 Offset press with 2 printing units.
- 01 Pallet inverter.
- 01 Control desk.
- 01 DELL PC.
- 01 Paper jogger.
- 01 HEIDELBERG Stitcher.
- 01 HEIDELBERG Perforator.

- 02 ORIGINAL PERFECTA paper cutters.
- 01 HEIDELBERG STAHFOLDER KH 78 folding machine.
- 01 MBO folding machine.
- 01 MULLER MARTINI production line (stitcher, three-side cutter, and wrapper).
- 01 KOLBUS production line (gathering machine, perfect binder, three-side cutter, and wrapper).

6.3 Packaging Workshop (Where Magma is Produced)

This workshop is responsible for producing packaging products, including the Magma box. It includes the machines used for printing, die-cutting, folding, and gluing.

Equipment

- 01 Offset press with 6 printing units including varnish: HEIDELBERG XL106.
- 01 Pallet lift.
- 01 Control desk.
- 01 HEIDELBERG Stitcher.
- 01 HEIDELBERG Perforator.
- 01 PERFECTA Premium Line 115 paper cutter.
- 01 HEIDELBERG TL-36 folding machine.
- 01 MK DIANA Easy 115 folding-gluing machine.
- 01 DELL PC.
- 01 Paper jogger.
- 01 MK DIANA Easy MATRIX 106 CS die-cutting machine.

6.4 Rotary Workshop (Newspapers)

The Rotary Workshop is responsible for high-speed newspaper printing.

Equipment

- 01 COMMUNITY rotary machine with 2 outputs.
- 01 GOSS UNIVERSAL rotary machine with 2 outputs.
- 02 CMYK towers + 02 monochrome towers.
- (B) 01 CMYK tower + 03 monochrome towers.

- (F1) 01 CMYK tower + 02 monochrome towers.
- (F1) 02 CMYK towers + 02 monochrome towers.
- Reel weight between 480-450 kg for a grammage of 45 g.

6.5 Various Equipment (Handling and Transport)

SIA has equipment for handling and transport work.

Equipment

- 05 Forklifts.
- 14 Light transport vehicles.

Heavy Transport Equipment

- 02 Semi-trailer trucks.
- 01 Truck 07 tons.
- 02 Trucks 02.5 tons.
- 01 Van.
- Personnel transport vehicles.

6.6 Tools

SIA uses some tools for measurement and checking during work.

Tools

- Palmer.
- Pantone guide.
- Magnifying glass (Compte-fil).
- pH meter.

The following figures show the layout of equipment in the main workshops of SIA.

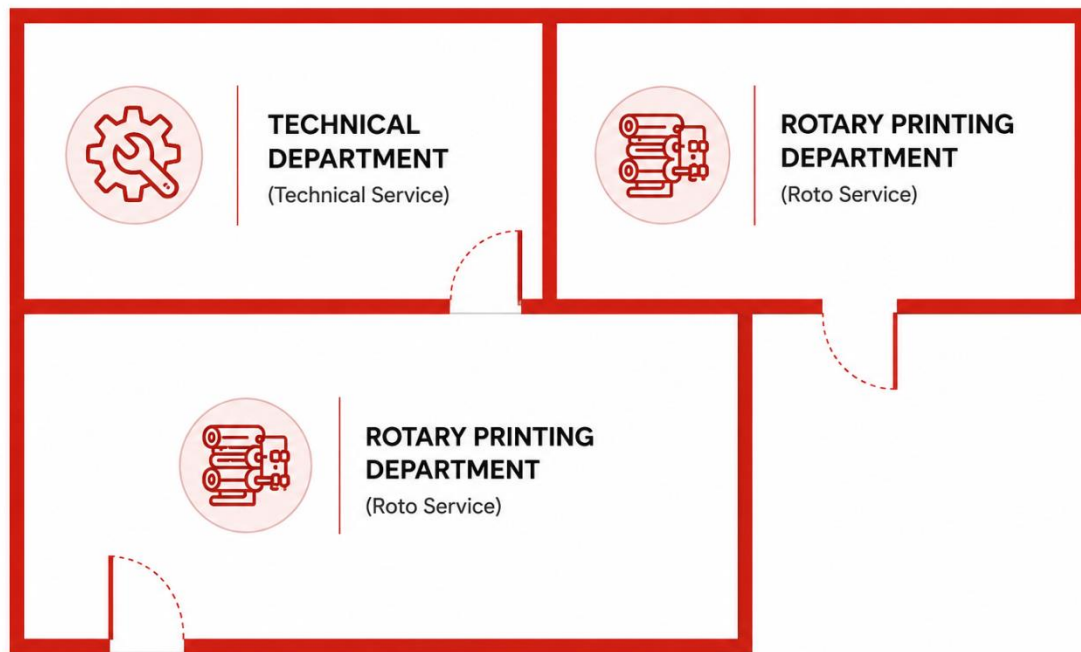


Figure 4 . Equipment Layout of the Rotary Workshop

Source: Designed by the author.

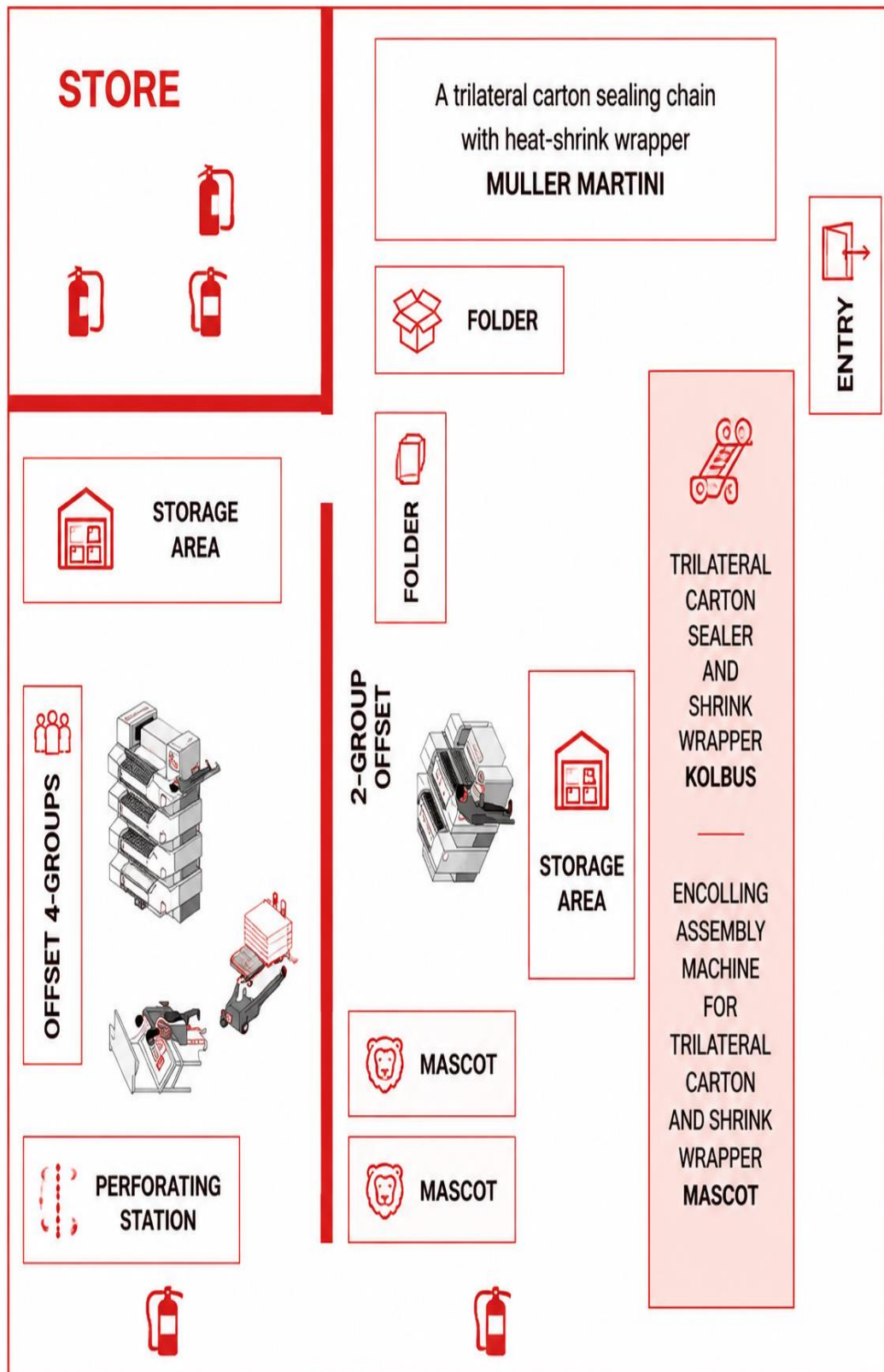


Figure 5. Layout of the Equipment in the Labeur Workshop

Source: Designed by the author.

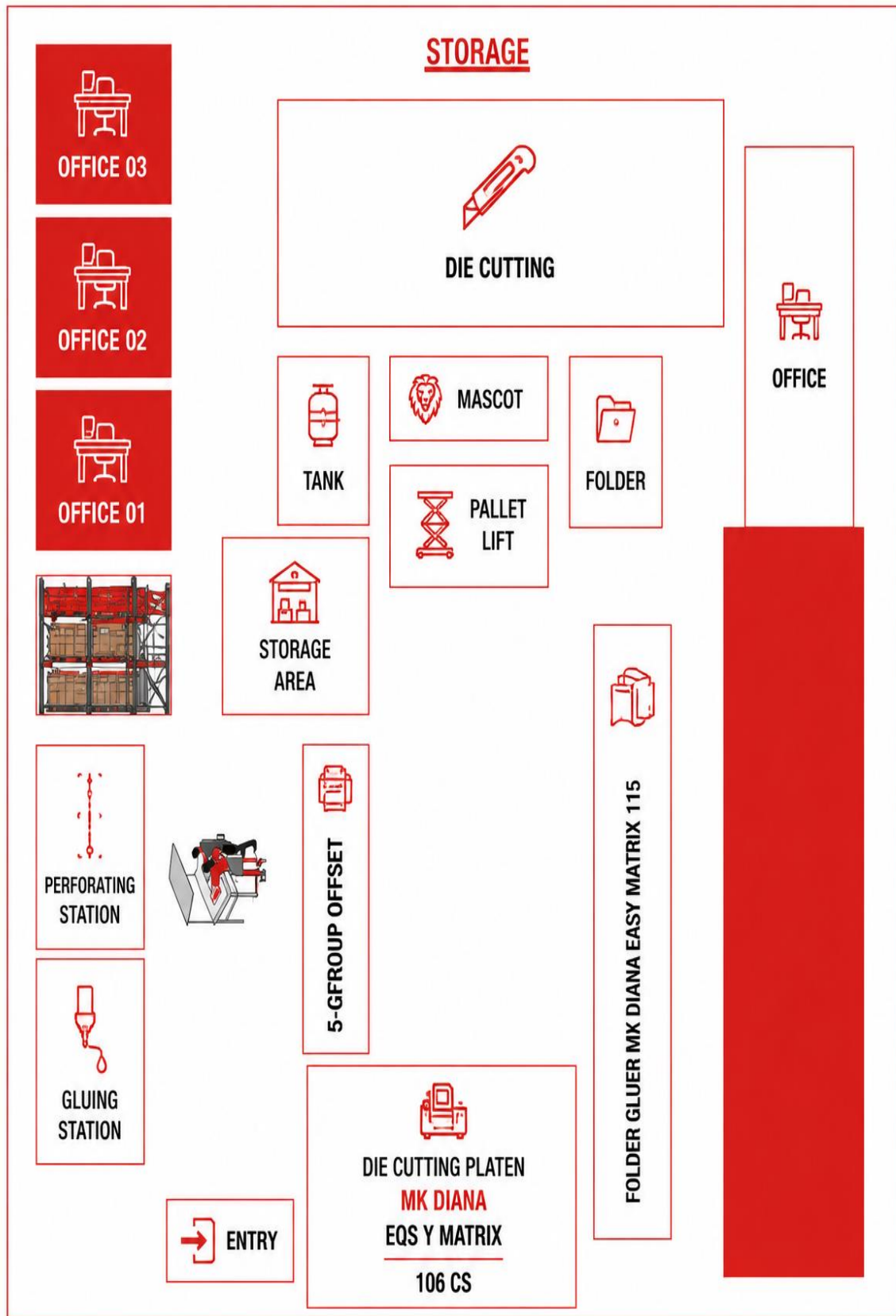


Figure 6. Layout of the Equipment in the Packaging Workshop

Source: Designed by the author.

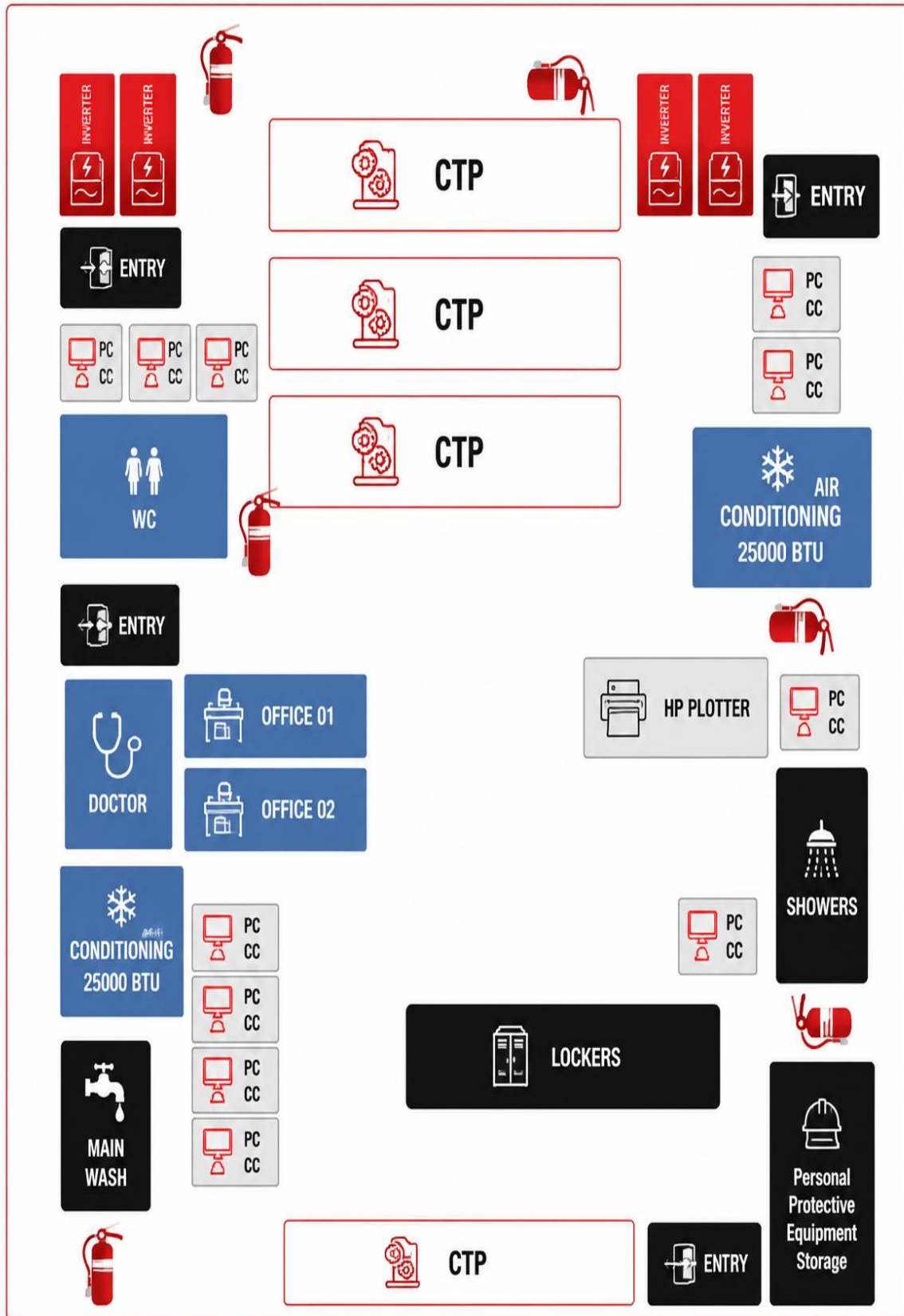


Figure 7. Layout of the Equipment in the Pre-press Workshop

Source: Designed by the author.

7. Main Clients of the Company

SIA works with clients from different business sectors. Some of its main clients are shown in the following figure.



Figure 8. Main Clients of the Company

Source: Internet.

8. Main Products of the Company

SIA produces different printed products for various industries. Some examples are presented in the following figure.



Figure 9. Main Products of the Company

Source: Internet.

Chapter Conclusion

This chapter presented SIA, its organization, services, equipment, clients, and main products.

It provides a general overview of the company and its printing and packaging activities.



CHAPTER N°2:

PROFESSIONAL IMMERSION

Introduction

During my internship at Société d'Impression d'Alger (SIA), I learned how a printing company works. I visited different workshops and followed the main steps of the printing process. This internship helped me improve my technical knowledge and understand how printed products are produced.

At the beginning of the internship, I spent some time observing the working methods used in the company. After that, I worked in different workshops under the supervision of the company's staff. This experience helped me understand the complete printing workflow, from pre-press to the final product.

1. Departments Visited During the Internship

During my internship, I worked in different workshops to learn the main steps of the printing process. Each workshop had a different role in the production line. I learned how the work starts with file preparation and ends with the final printed product.

The workshops I visited were:

- Pre-press Workshop
- Press Workshop
- Post-press Workshop



Figure 10. Pre-press Workshop

Source: Taken by the author.

1.1. Pre-press Workshop

In the Pre-press Workshop, I learned how printing files are prepared before printing. The operators checked the customer files and corrected them when needed. They also prepared the printing plates and printed the digital proof (BAT) before production.

During this period, I participated in the following activities:

- Checking customer files.
- Measuring the printing margins.
- Checking text overprint and color registration.
- Preparing printing plates using the CTP system.
- Printing the BAT.
- Helping with the replacement of CTP chemicals.
- Working with Adobe Illustrator, Adobe Acrobat Pro, and Kodak PREPS.
- Organizing the printing plates before production.

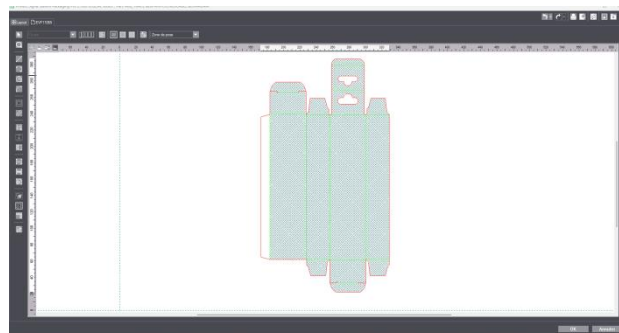


Figure 11. Checking and Preparing Customer Files

Source: Taken by the author.



Figure 12. Computer-to-Plate (CTP) System

Source: Taken by the author.

1.2. Press Workshop

In the Press Workshop, I learned how the printing machines are prepared before production. I followed the different machine settings and observed the printing process until the final printed sheets were produced.

During this period, I participated in the following activities:

- Reading the production file.
- Preparing the printing machine.
- Loading the paper.
- Installing the printing plates.
- Adjusting the ink system.
- Adjusting the dampening system.
- Adjusting the feeder.
- Adjusting the color registration.
- Starting the production process.
- Applying acrylic varnish after printing.

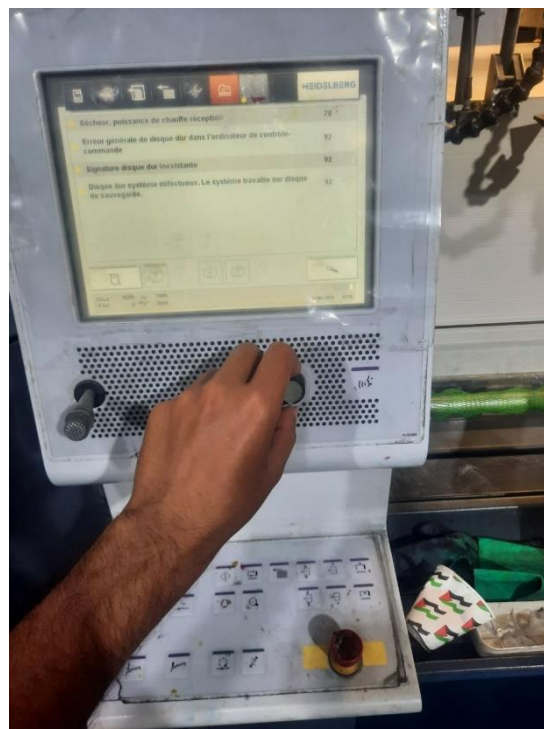


Figure 13. Control Panel of the Heidelberg XL106

Source: Taken by the author.



Figure 14. Heidelberg XL106

Source: Taken by the author.

1.3. Post-press Workshop

In the Post-press Workshop, I learned the final steps of packaging production. After printing, the sheets were cut, folded, glued, and prepared to become finished packaging boxes. I observed how the operators adjusted each machine before starting production and checked the quality of the final boxes.

During this period, I participated in the following activities:

- Preparing the die-cutting machine.
- Adjusting the feeder.
- Adjusting the cutting pressure.
- Adjusting the delivery section.
- Starting the die-cutting process.
- Removing the waste from the die-cut sheets.
- Preparing the folding and gluing machine.
- Adjusting the glue system.
- Adjusting the folding guides.
- Adjusting the delivery section.
- Producing the final packaging boxes.



Figure 15. Die-cutting Process

Source: Taken by the author.



Figure 16. MK Easymatrix 106 CS Die-cutting Machine

Source: Taken by the author.



Figure 17. MK Diana Easy 115 Folding and Gluing Machine

Source: Taken by the author.



Figure 18. Finished Packaging Boxes

Source: Taken by the author.

2. Tasks and Missions

During my internship, I participated in different technical tasks in the three workshops. These tasks helped me understand the complete production process, from preparing the printing files to producing the final packaging boxes.

2.1. File Verification

The first step was checking the customer's files before production. The files were verified using graphic software to make sure that the layout, colors, and printing margins were correct.

The following points were checked:

- File dimensions.
- Bleed and safety margins.
- Color registration.
- Overprint settings.
- Barcode position.
- General layout.

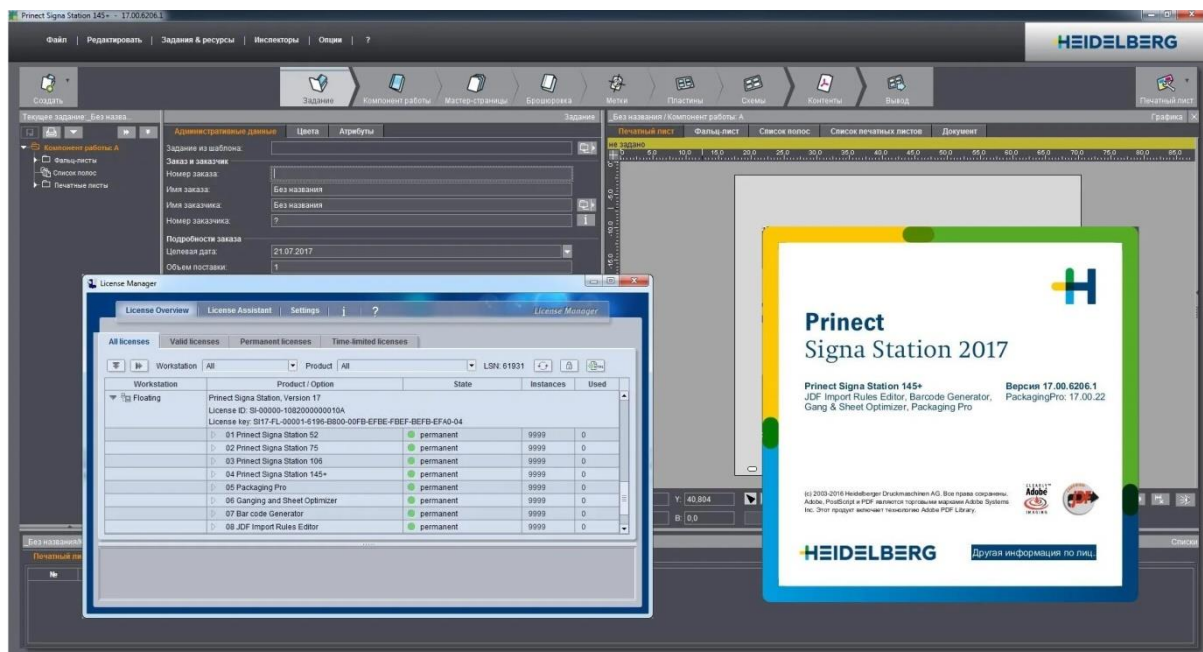


Figure 19. File Verification in Prinect Package Designer

Source: Taken by the author.

2.2. Plate Production

After the files were checked, the printing plates were made using the CTP system. I learned how the plates were prepared before printing.

During this work, I:

- Checked the PDF files.
- Sent the files to the CTP machine.
- Watched the plate production.
- Prepared the plates for printing.

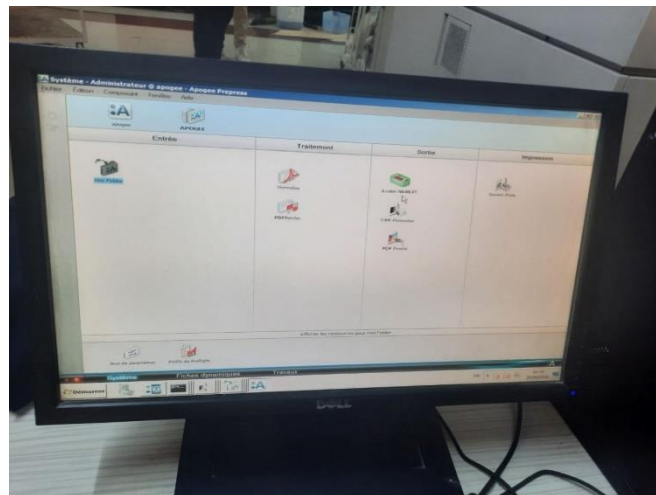


Figure 20. Screenshot of the CTP software

Source: Taken by the author.



Figure 21. Printing Plates

Source: Taken by the author.

2.3. Quality Control

During my internship, I also learned how quality control is done during printing. I followed the quality control work on different printed products, such as a School Communication Book. The operators checked the printed sheets during production to make sure that the product had good quality.

They checked:

- The printing colors.
- The print quality.
- Printing defects.
- The varnish.
- The cutting and folding quality.
- The final appearance of the product.

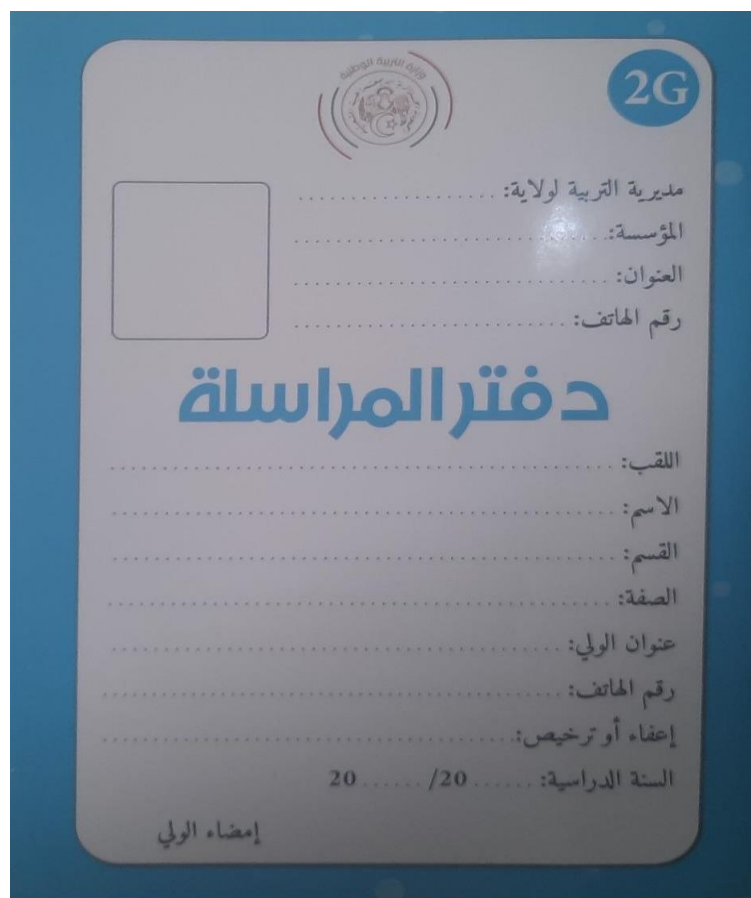


Figure 22. Finished School Communication Book

Source: Taken by the author.

3. Suggestions

During my internship, I found some ideas that may help improve the work in the company.

To improve working conditions

- Organize the storage area better.
- Provide more carts to move materials between the workshops.
- Improve the lighting in some working areas.

To improve production

- Prepare all production materials before starting the job.
- Continue regular machine maintenance to reduce machine stops.
- Continue checking the quality during all production stages.

To improve organization

- Improve communication between the workshops.
- Organize the production documents better.
- Improve cooperation between the Pre-press, Press, and Post-press workshops.

Chapter Conclusion

This internship allowed me to understand the complete offset printing workflow. I learned how printing files are prepared, how the Heidelberg Speedmaster XL 106 is adjusted before production, and how printed sheets are converted into finished packaging boxes using die-cutting, folding, and gluing machines.

I also improved my practical skills by participating in several production tasks and by observing quality control during printing. This experience helped me better understand industrial printing and prepared me for the realization of my graduation project.



PART II :

PROFESSIONAL PROJECT



CHAPTER N°3 : PROJECT PREPARATION

Introduction

This chapter presents the study and production of the Magma Printer's Cable Packaging Box. It covers the project objectives, production workflow, technical specifications, production cost, timeline, and manufacturing file.

1. Project Presentation

This project is about the production of the Magma Printer's Cable Packaging Box. The packaging was produced using the offset printing process at the Packaging Unit of Société d'Impression d'Alger (SIA).

Before starting production, the project had to be prepared carefully. The customer's requirements were checked first. Then, the technical specifications of the box were defined. After that, the production process was planned before the work started.

This preparation helped all departments work together and follow the same production plan.

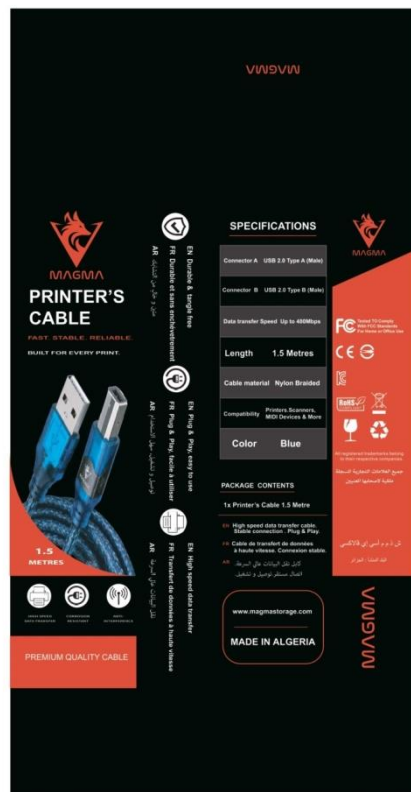


Figure 23. Magma Printer's Cable Packaging Box

Source: Taken by the author.

2. Project Objectives

The main objective of this project was to produce the Magma Printer's Cable Packaging Box using the offset printing process.

This project also helped me:

- To prepare the production files.
- To follow the production process.
- To estimate the production cost.
- To organize the production work.
- To produce a packaging box that meets the customer's requirements.

3. Production Workflow

The production of the Magma packaging box followed several steps. Each department completed its work before the next department started. This workflow helped organize the production process and ensured that the packaging was produced correctly.

The production workflow was:

- Customer
- Commercial Department
- Pre-press Workshop
- Offset Printing
- Die-cutting
- Folding and Gluing
- Packaging

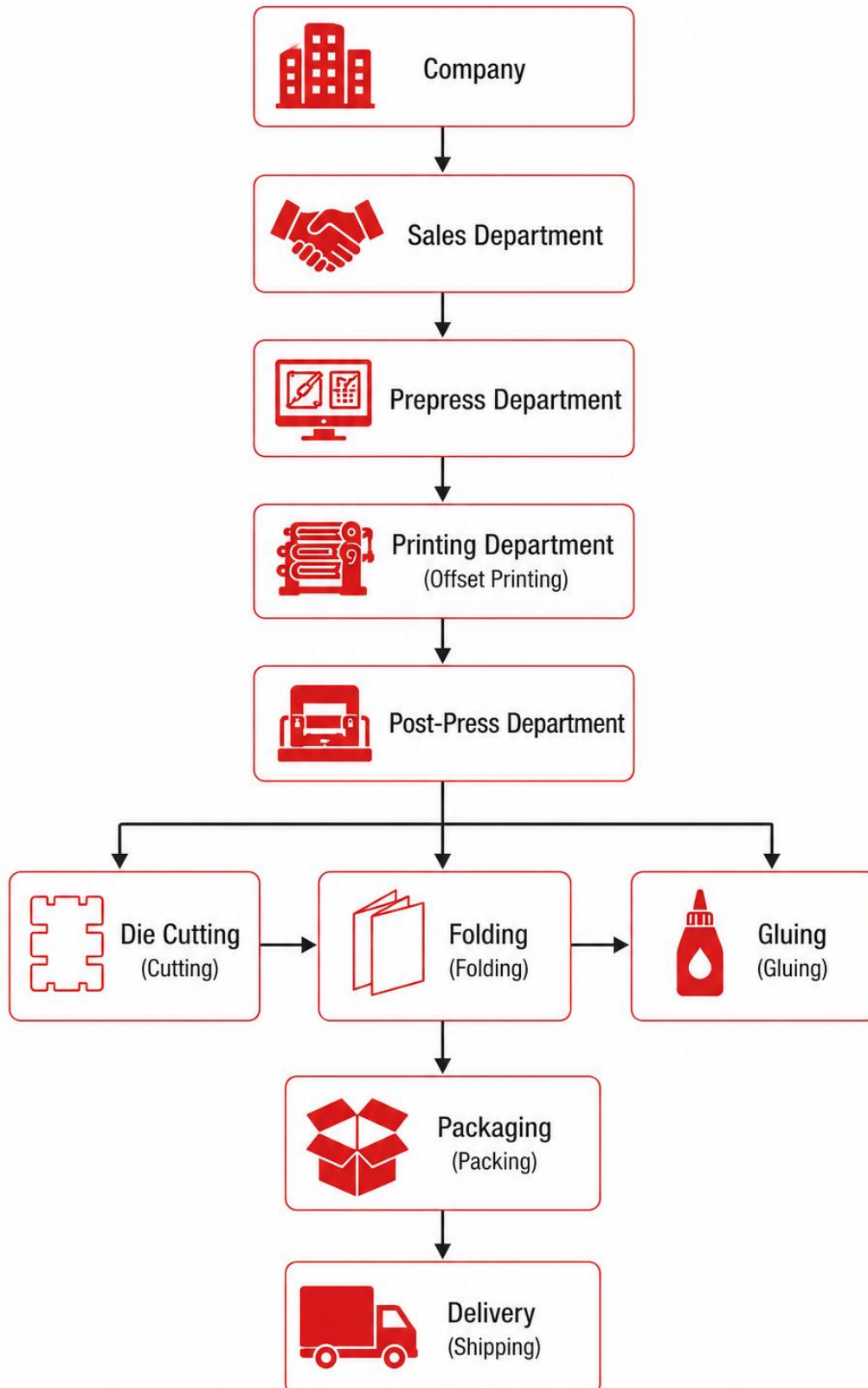


Figure 24. Production Workflow of the Magma Packaging Box

Source: Designed by the author

4. Technical Specifications

Item	Specification
Product	Magma Printer's Cable Packaging Box
Client	SARL ACI GALAXY
Packaging Style	Reverse Regular Tuck Lock with Hanging Panel (EVF11065)
Finished Size	44 × 29 × 177 mm
Printing Process	Offset Printing
Printing Machine	Heidelberg Speedmaster XL 106
Printing Colors	Cyan, Yellow, Black and Pantone 485 C + Pantone 485 C
Paper Type	BBTB Cardboard
Grammage	300 g/m ²
Surface Finish	Acrylic Matt Varnish
Printing Sheet	700 × 1000 mm Printing Sheet
Bleed	3 mm
Production Quantity	20,000 Boxes

Table 1. Technical Specifications

5. Manufacturing Cost Calculation

PRODUCTION COST CALCULATION

Study and Production of the Magma Printer's Cable Packaging Box

5.1.General Project Information

This section presents the actual production cost calculation of the Magma Printer's Cable Packaging Box. The calculation is based on the real production data used during the manufacturing process at SIA.

Parameter	Value
Order quantity	20,000 units
Printing process	Offset Printing — 4 colors + Acrylic Varnish
Packaging material	BBTB Cardboard, 300 g/m ²

5.2.Technical Specifications

Parameter	Value
Quantity	20,000 units
Imposition	12 units per sheet
Paper	BBTB Cardboard, 300 g/m ²
Sheet format	700 × 1,000 mm (0.70 m ²)
Printing process	Offset Printing, 4 colors (Cyan, Yellow, Black and Pantone 485 C) + Acrylic Varnish

Parameter	Value
Waste allowance	9%

5.3.Production Parameters

This section shows the production information. These production values were used to calculate the actual production cost of the project.

Calculation step	Result
Useful sheets required ($20,000 \div 12$)	$1,666.67 \approx 1,667$ sheets
Waste allowance (9% of 1,667)	≈ 150 sheets
Total sheets required	1,817 sheets
Sheet surface area ($0.70 \text{ m} \times 1.00 \text{ m}$)	0.70 m^2
Weight per sheet ($0.70 \text{ m}^2 \times 300 \text{ g/m}^2$)	210 g
Total paper weight ($1,817 \times 210 \text{ g}$)	381,570 g = 381.57 kg

5.4.Material Cost – Paper

This section shows the cost of the paper used for the project. The paper cost was calculated using the actual paper weight and the paper unit price.

Calculation	Result
$381.57 \text{ kg} \times 250 \text{ DA/kg}$	95,392.50 DA

Paper Cost	95,392.50 DA
-------------------	---------------------

5.5. Plate and Tooling Cost

CTP Plates

Four CTP plates were prepared for this project. One plate was prepared for each printing color.

Calculation	Result
$4 \times 1,500 \text{ DA}$	6,000 DA

CTP Plates Cost	6,000.00 DA
------------------------	--------------------

Die-Cutting Mold

This die-cutting mold was used to cut and crease the packaging box before folding and gluing.

Calculation	Result
$1 \times 5,000 \text{ DA}$	5,000 DA

Die-Cutting Mold Cost	5,000.00 DA
------------------------------	--------------------

5.6. Ink Cost

This section shows the ink quantity used during production.

Color	Quantity (kg)
Cyan	1.0

Color	Quantity (kg)
Yellow	1.0
Black	0.8
Pantone 485 C	0.7
Total	3.5

Ink unit price: 4,500 DA/kg

Calculation	Result
$3.5 \text{ kg} \times 4,500 \text{ DA/kg}$	15,750 DA

Ink Cost	15,750.00 DA
-----------------	---------------------

5.7.Varnish Cost

This section shows the acrylic varnish consumption used for the project.

Calculation step	Result
Total printed surface ($1,817 \times 0.70 \text{ m}^2$)	1,271.90 m ²
Total varnish consumption ($1,271.90 \text{ m}^2 \times 6 \text{ ml/m}^2$)	7,631.40 ml $\approx$ 7.63 L
Cost ($7.63 \text{ L} \times 420 \text{ DA/L}$)	3,204.60 DA

Varnish Cost	3,204.60 DA
---------------------	--------------------

5.8.Glue Cost

This section shows the glue used for the project. The glue quantity used for this project was 6 kg, with a unit price of 700 DA/kg.

Calculation	Result
6 kg × 700 DA/kg	4,200 DA

Glue Cost	4,200.00 DA
------------------	--------------------

5.9.Labour Cost

This section shows the labour cost assigned to this production order.

Labour Cost	10,000.00 DA
--------------------	---------------------

5.10.Other Charges

This section shows the other production costs. A 2% overhead charge was applied according to the company's production cost calculation.

Calculation	Result
139,547.10 DA × 2%	2,790.94 DA

Overhead Charges	2,790.94 DA
-------------------------	--------------------

5.11.Production Cost Summary

This section shows the total production cost. The table below summarizes the actual production costs of the project.

Cost Item	Amount (DA)
Paper	95,392.50
Varnish	3,204.60
Inks	15,750.00
CTP Plates	6,000.00
Die-Cutting Mold	5,000.00
Glue	4,200.00
Subtotal	129,547.10
Labour	10,000.00
Total production cost	139,547.10
Overhead charges (2%)	2,790.94

Total Production Cost, Excl. Tax (HT)	142,338.04 DA
--	----------------------

5.12. Unit Cost

This section shows the unit cost of the project. The unit cost was obtained by dividing the total production cost by the production quantity.

Calculation	Result
142,338.04 DA ÷ 20,000	7.12 DA

Unit Cost Price (HT)	7.12 DA
-----------------------------	----------------

5.13.Selling Price

This section shows the selling price of the project. The selling price was obtained by adding a commercial margin of 25% to the unit cost.

Calculation	Result
$7.12 \times 25\%$	1.78 DA

Unit Selling Price (7.12 + 1.78)	8.90 DA
---	----------------

Total Sales Value

This section shows the total sales value of the project. It is obtained from the order quantity and the unit selling price. After that, 19% VAT (TVA) is added to obtain the total price.

Description	Quantity	Unit Price HT (DA)	Amount HT (DA)
Magma Printer's Cable Packaging Box	20,000	8.90	178,000.00

Item	Amount (DA)
Total, Excl. Tax (HT)	178,000.00
VAT / TVA (19%)	33,820.00

Item	Amount (DA)
Total, Incl. Tax (TTC)	211,820.00

Note: All prices are in Algerian Dinar (DA). HT: Excluding Tax • TTC: Including Tax • VAT (TVA): Value Added Tax.

4. Production Timeline

This section shows the production timeline of the Magma Printer's Cable Packaging Box. The table below shows the production tasks and the time required for each task

Task	Code	Previous Task	Duration
Manufacturing Study and Cost Estimation	A	—	2 h
Customer File Verification	B	—	0.5 h
Imposition and Layout Preparation	C	B	2 h
CTP Plate Production	D	C	0.5 h
Offset Press Setup	E	D	1 h
Offset Printing	F	E	1 h
Die-Cutting Machine Setup	G	F	1.5 h
Die-Cutting Operation	H	G	1 h
Waste Removal	I	H	45 min
Folder-Gluer Setup	J	I	30 min

Folding and Gluing	K	J	1.5 h
Final Packaging and Carton Arrangement	L	K	30 min

Table 2. Production schedule of the Magma Printer's Cable Packaging Box

Production Workflow Diagram

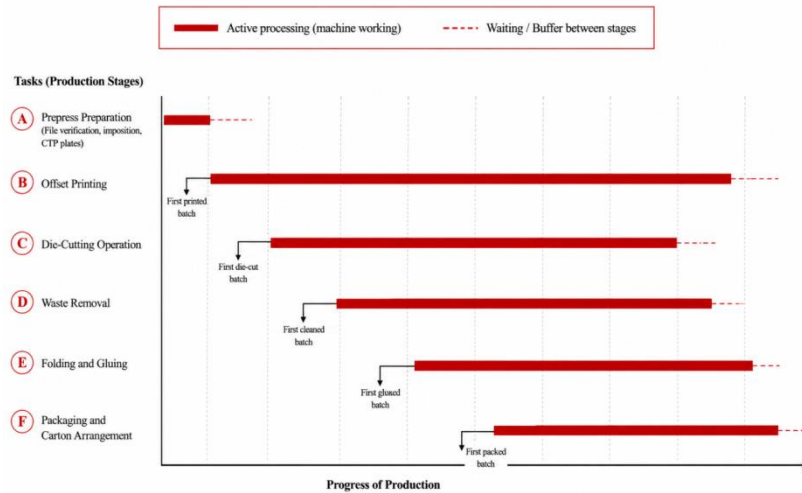


Figure 25. Parallel Production Workflow of the Magma Printer's Cable Packaging Box

Source: Designed by the author

Analysis

This diagram shows the production process of the Magma Printer's Cable Packaging Box.

The machines did not work one by one. They worked together. When one batch was ready, it moved to the next production stage. The previous machine continued working on the remaining batches.

This production method helped save time and improved the work.

The production order was completed in three working days under the normal working schedule.

5. Manufacturing File

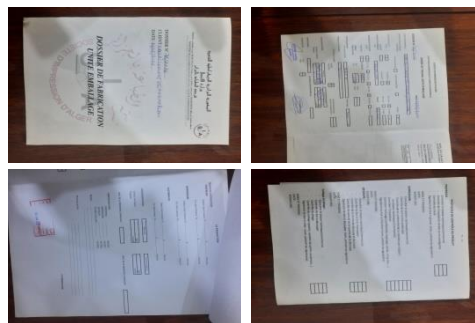
The manufacturing file is an internal document used by SIA during the production of the Magma Printer's Cable Packaging Box.

This file contains all the technical and production information needed to manufacture the product. It follows the job from the beginning of production to the final packaging stage. Each department uses this file and records its work before sending the job to the next production stage.

The manufacturing file used in this project includes:

- Work order number and production date.
- Customer information.
- Product description.
- Finished box dimensions.
- Printing specifications.
- Paper type and grammage.
- Printing colors and varnish.
- Die-cutting and finishing information.
- Production records.
- Quality control and validation forms for each production department.

The following figures show the manufacturing file used during the production of the Magma Printer's Cable Packaging Box.



Chapter Conclusion

This chapter presented the main steps of studying and producing the Magma packaging box, from preparation to final production. It also covered the technical, cost, time, and manufacturing aspects of the project.



CHAPTER N°4 : PROJECT IMPLEMENTATION



SECTION 01 : PREPRESS

Introduction

This chapter presents the prepress and plate-making stages of the Magma Printer's Cable Packaging Box. It explains the main steps used to prepare the artwork, create the packaging structure, prepare the B.A.T, and produce and check the printing plates.

1. Prepress Department

The prepress department is the first production stage before printing. Its main job is to prepare and check the digital files to make sure they are ready for printing. During this stage, every file is verified to avoid printing errors and to ensure that the final packaging meets the required quality standards.

The software used during the prepress stage includes:

- Adobe Acrobat Pro DC
- Adobe Illustrator
- Prinect Package Designer
- Prinect Signa Station

2. Prepress Workflow

The following flowchart presents the main workflow followed in the prepress department during the production of the Magma Printer's Cable Packaging Box.

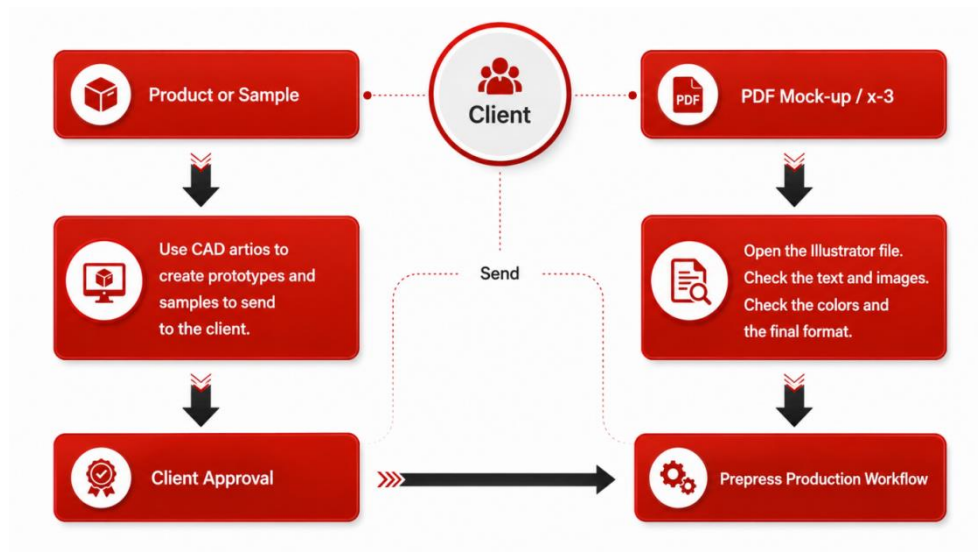


Figure 27. Prepress workflow of the Magma Printer's Cable Packaging Box

Source: Designed by the author

3. Receiving the Customer File

The production process starts when the customer sends the final artwork in PDF format. This file contains all graphic elements of the packaging, including the logo, product information, colours, images, and technical specifications. Before production, the PDF file is carefully inspected to ensure that it complies with the printing requirements.

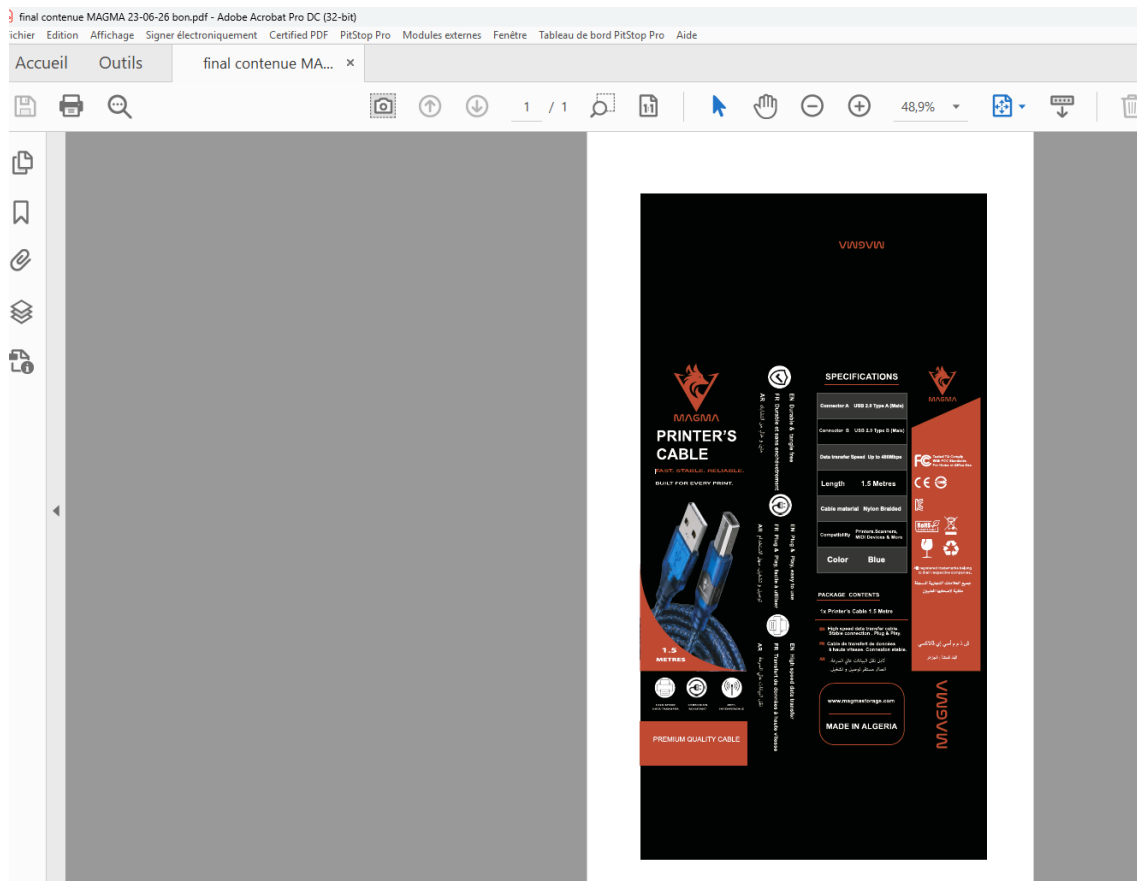


Figure 28. Final PDF artwork received from the customer

Source: Taken by the author.

4. Verification and Processing of the PDF File

After receiving the PDF file from the customer, the file is opened in Adobe Illustrator. Before starting the packaging design, we check the file carefully to make sure that everything is correct. We also import the die-cut template to make a first verification and to see if the graphic design fits correctly inside the cutting lines.

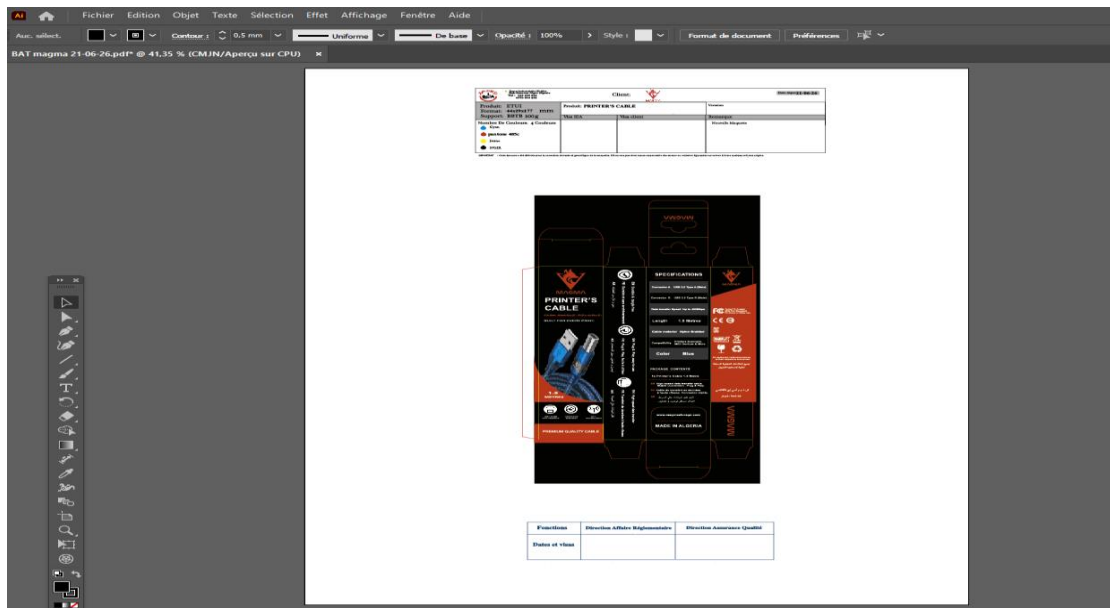


Figure 29. Opening the PDF file in Adobe Illustrator and checking the compatibility with the die-cut template.

Source: Taken by the author.

During this step, the following points are checked:

- The file is saved in PDF/X-4 format.
- The final size of the packaging is checked.
- Bleed and safety margins are checked.
- All fonts are embedded or converted to outlines.
- The document is prepared in CMYK color mode.
- The Pantone 485 C spot color is correctly defined.
- The barcode is 100% Black.
- Image resolution is 300 dpi.
- Graphic layers, die-cut lines and printing marks are separated correctly.
- The graphic design is checked with the die-cut template to make sure that all texts, logos and images are inside the correct position and no important element will be cut during production.

After all these checks, the PDF file is exported according to the production standards used by Société d'Impression d'Alger (SIA).

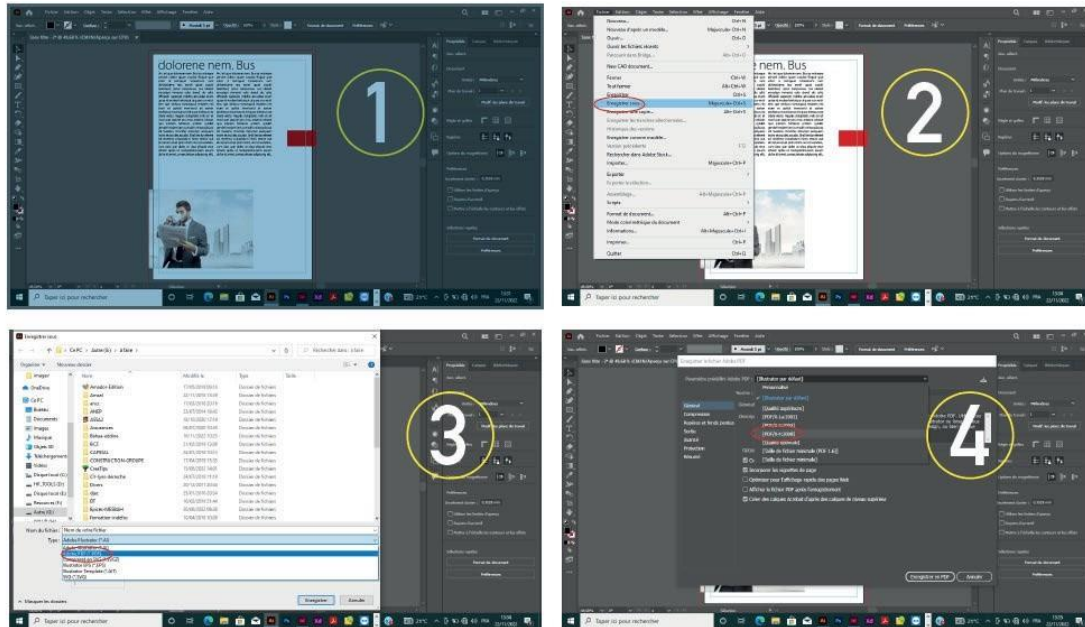


Figure 30. PDF/X-4 export according to SIA production standards

Source: SIA company.

The export is done by choosing the PDF/X-4 profile recommended by SIA. This file will be used in the next production steps.

5. Structural Design of the Packaging

The packaging structure was made with Prinect Package Designer. This software is used to create the box shape, change the dimensions, check the folding, and prepare the packaging before printing and die-cutting. The work was done step by step to make sure that the packaging was correct before production.

5.1 Selection of the Packaging Template

The first step was to choose a packaging template from the software library. We searched for a box similar to our product, then we selected the template EVF11065 – Reverse Regular Tuck Lock with Hanging Panel. This template was suitable for the Magma Printer's Cable Packaging Box because it has the correct box structure and a hanging panel for product

display. Before starting the design, we checked the template information and the 3D preview to make sure that it was the correct model for our project.

Princt Package Designer

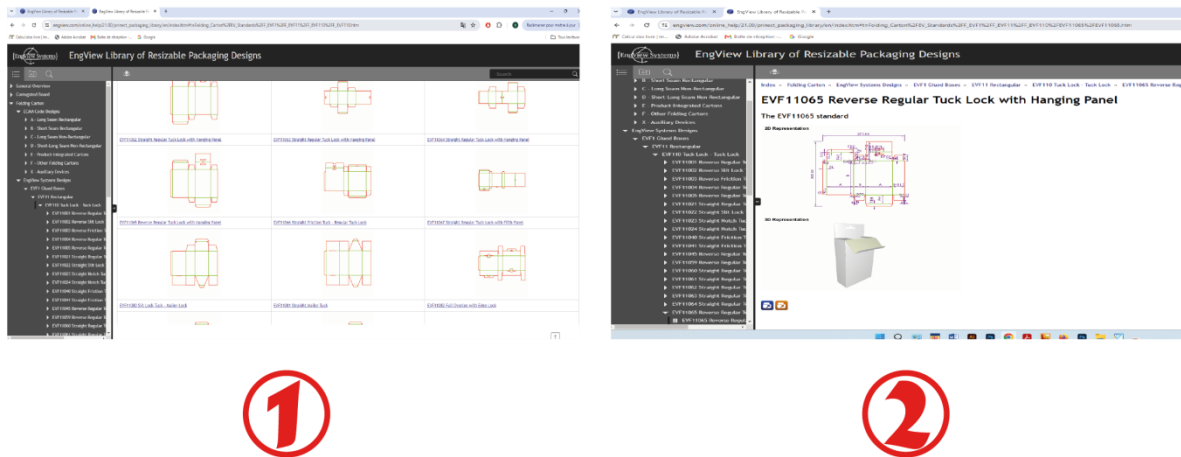


Figure 31. Selection of the Packaging Template (EVF11065)

Source: Taken by the author.

5.2 Adjustment of the Packaging Dimensions

After selecting the template, the box dimensions were changed according to the specifications of the Magma packaging. The length, width, height, and hanging panel dimensions were entered into the software. Once the new values were inserted, the software updated the complete box automatically. This allowed us to keep all structural parts connected correctly without redrawing the package.

Prinect Package Designer

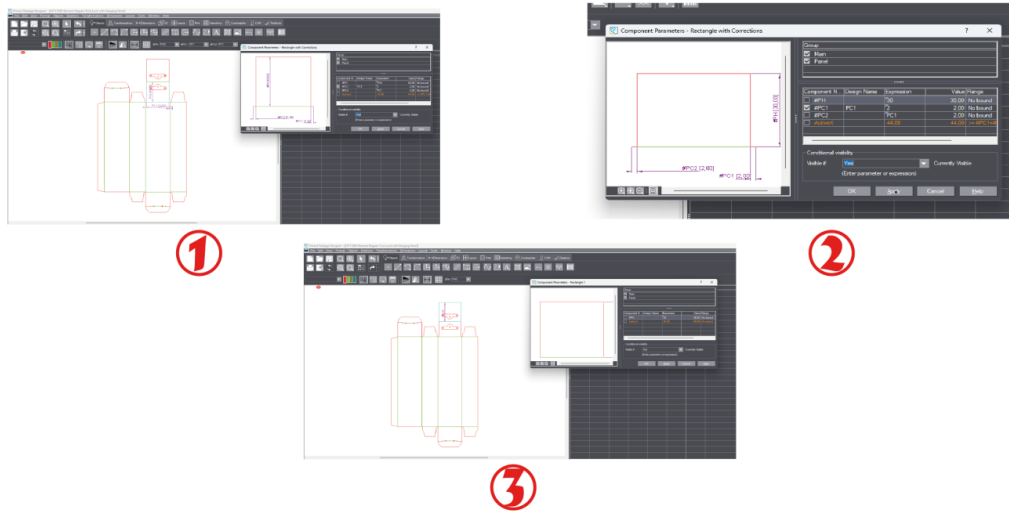


Figure 32. Adjustment of the Packaging Dimensions

Source: Taken by the author.

5.3 Adjustment of the Hanging Hole and Cutting Lines

The hanging hole was adjusted to fit the product display requirements. The rounded corners and cutting lines were also modified to obtain a clean die-cutting result and to ensure that the box could be folded correctly without damaging the cardboard during production.

Prinect Package Designer

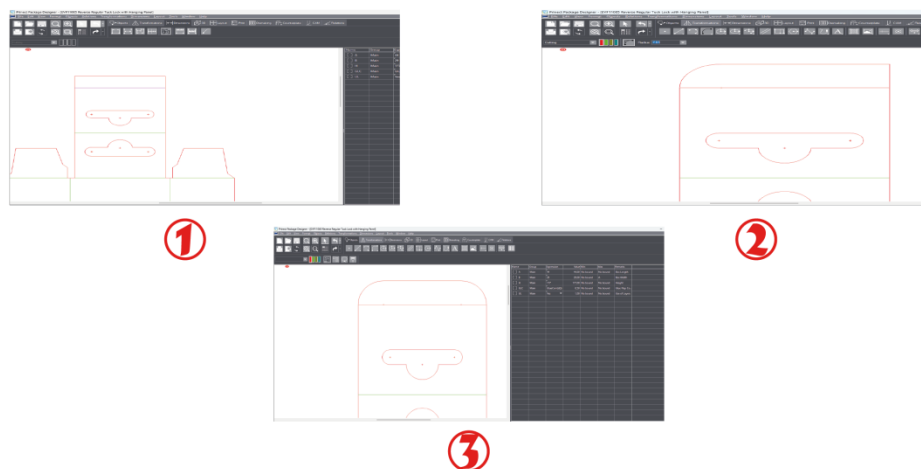


Figure 33. Adjustment of the Hanging Hole and Cutting Lines

Source: Taken by the author.

5.4 Positioning the Artwork

After finishing the box structure, the artwork was placed on the packaging layout. The logo, product name, images, barcode, and technical information were positioned on each panel of the box. We checked that all elements were inside the safe area and away from the cutting and folding lines. This step helped us make sure that the printed design would appear correctly after the box was folded.

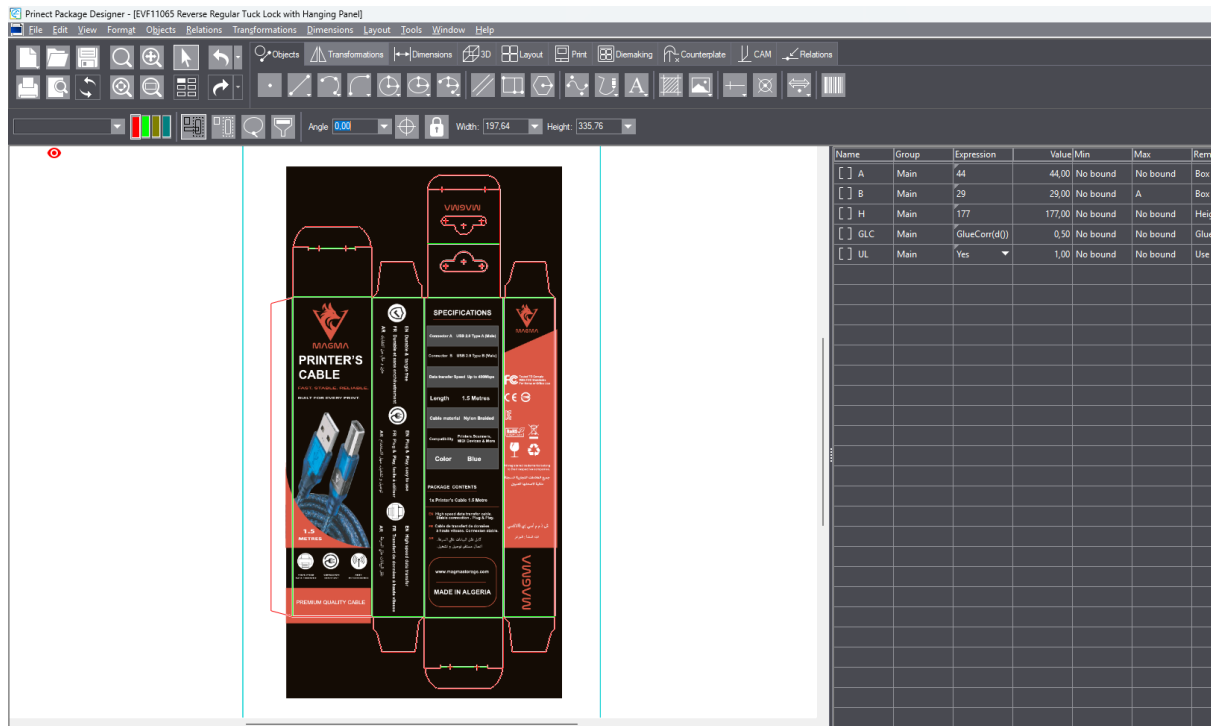


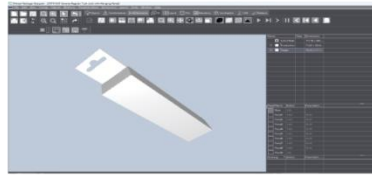
Figure 34. Positioning the Artwork on the Packaging Layout

Source: Taken by the author.

5.5 Three-Dimensional Visualization

The packaging was converted into a 3D model to check the final shape. The box was folded inside the software to verify that all panels closed correctly. We also checked the position of the printed design on every side of the box. This preview helped us find possible problems before production.

Prinect Package Designer



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③



④

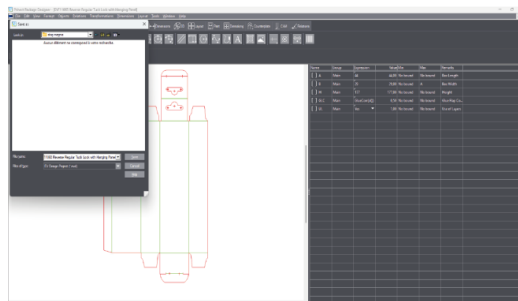
Figure 35. Three-Dimensional Visualization of the Packaging

Source: Taken by the author.

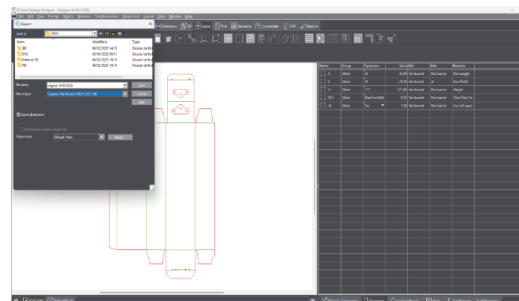
5.6 Saving and Exporting the Project

After all checks were completed, the project was saved in the original EVD format. This file can be opened later if any modification is needed. Then the die-cut file was exported in CFF2 format. This format is used in the next production step to prepare the imposition and the cutting process.

Prinect Package Designer



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Figure 36. Saving and Exporting the Packaging Project

Source: Taken by the author.

6. Elaboration of the B.A.T

After finishing the structural design of the packaging box, the next step was to prepare the B.A.T using Prinect Signa Station. This software was used to create the production job, prepare the printing layout, place the packaging on the printing sheet, and generate the final production file.

6.1 Creating a New Job

The first step was to create a new job in Prinect Signa Station. We clicked Create a New Job and entered the customer information, the product name, and the production quantity. Then, we selected the packaging option and prepared the workspace. This step creates the production job and prepares the software before importing the die-cut file.

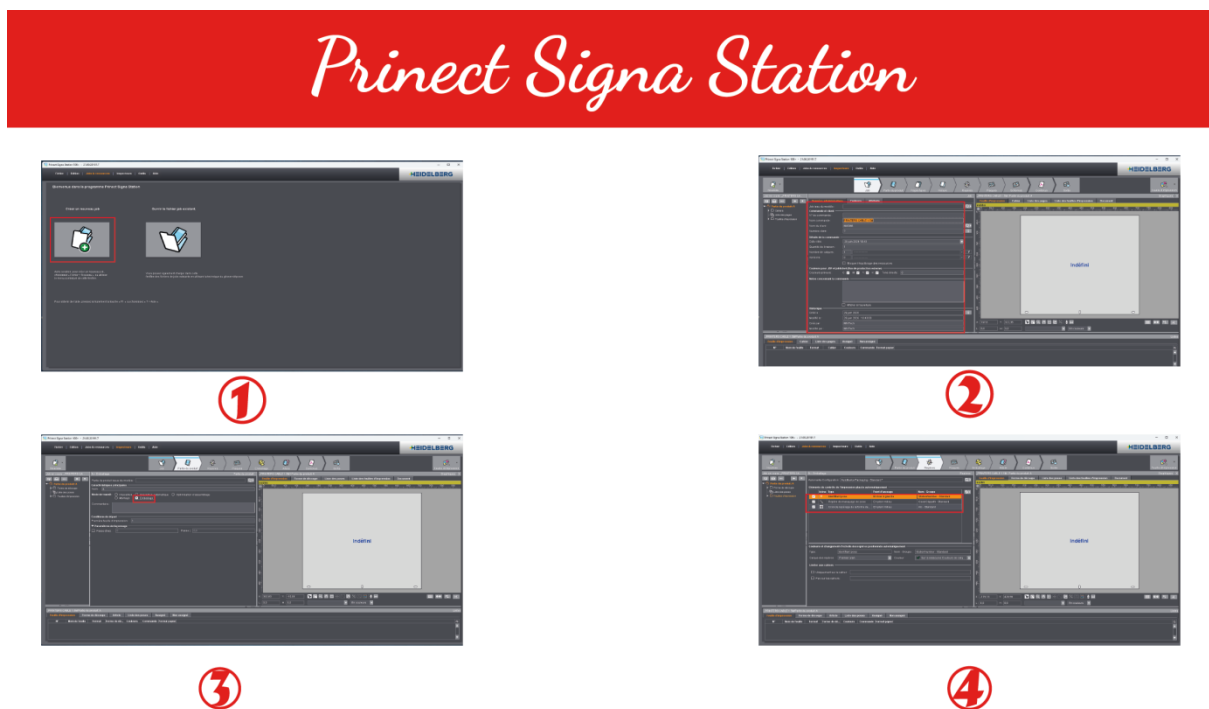


Figure 37. Creating a new production job and preparing the workspace in Prinect Signa Station
Source: Taken by the author.

6.2 Importing the Die-Cut File

After creating the new job, the die-cut file was imported into the project. The correct packaging design was selected from the list and loaded into the workspace. Then, the die-cut appeared inside the working area.

This step is important because it gives the software the real shape of the box. Without the die-cut file, it is impossible to prepare the printing layout correctly. It also allows the operator to continue all the next production steps.

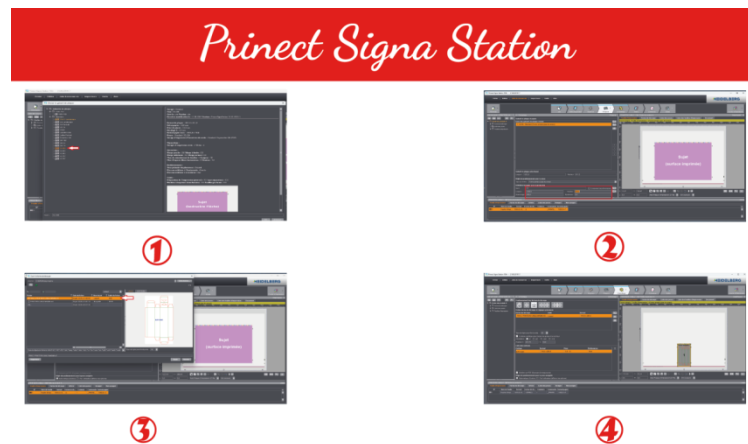


Figure 38. Importing the die-cut file into Prinect Signa Station and loading it into the project

Source: Taken by the author.

6.3 Setting the Packaging Layout

The packaging was placed on the printing sheet. The position and the direction of the box were adjusted before making copies. The software showed the printable area and helped place the box correctly.

This step is important because the box must be in the correct position before making the printing layout. A good position helps avoid printing and cutting problems.

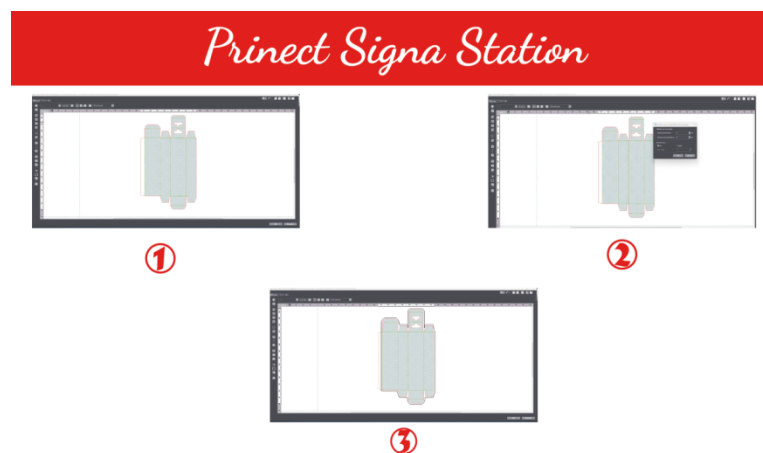


Figure 39. Setting the first packaging layout in Prinect Signa Station

Source: Taken by the author.

6.4 Creating the Step and Repeat Layout

After the first box was ready, the software made more copies of the box on the sheet. We changed the horizontal distance and the vertical distance between the boxes. We also checked the position until all boxes were correct on the sheet.

This step is important because it saves paper. It also gives more boxes in one printing sheet and prepares the file for printing.

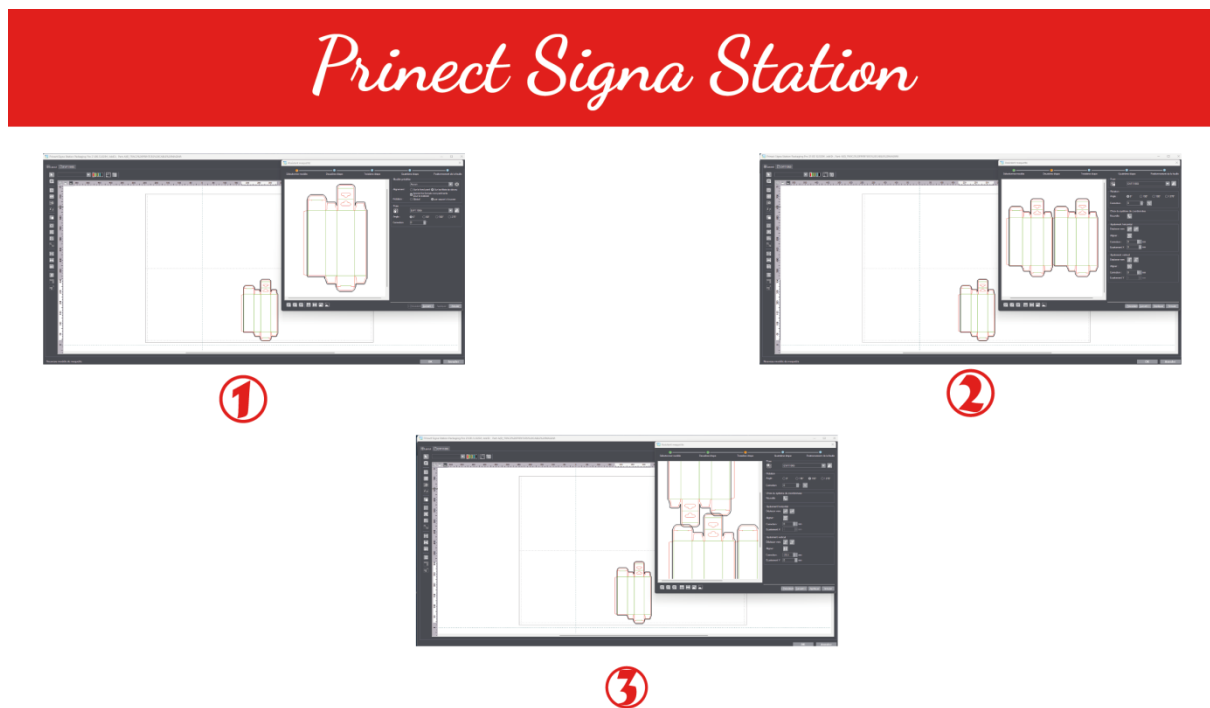


Figure 40. Creating the printing layout for the packaging

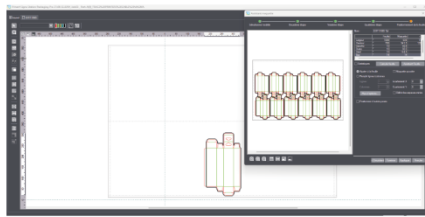
Source: Taken by the author.

6.5 Importing the Artwork

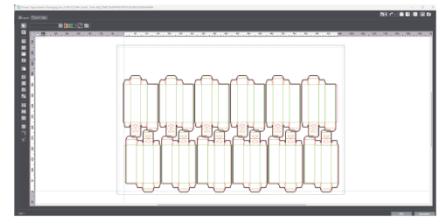
The final PDF file was imported into the project. The graphic design was linked with the die-cut layout. After that, the software showed the complete packaging with the artwork.

This step is important because the printed design must match the die-cut exactly. A correct match prevents printing outside the cutting lines.

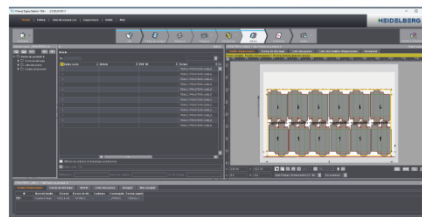
Prinect Signa Station



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③

Figure 41. Importing the final artwork into the packaging layout

Source: Taken by the author.

6.6 Adjusting the Artwork Position

We adjusted the artwork position on the die-cut. We changed the horizontal position and the vertical position until the design was correct on every side of the box.

This step is important because a wrong position can make the print go outside the cutting line. A correct position gives a good final box.

Prinect Signa Station



Figure 42. Adjusting the artwork position on the die-cut

Source: Taken by the author.

6.7 Final Layout Verification

The complete layout was checked before production. The box positions, the artwork, and the printing marks were verified. Every copy on the sheet was inspected carefully.

This step is important because it helps find mistakes before printing. It saves time, paper, and production cost.

Prinect Signa Station

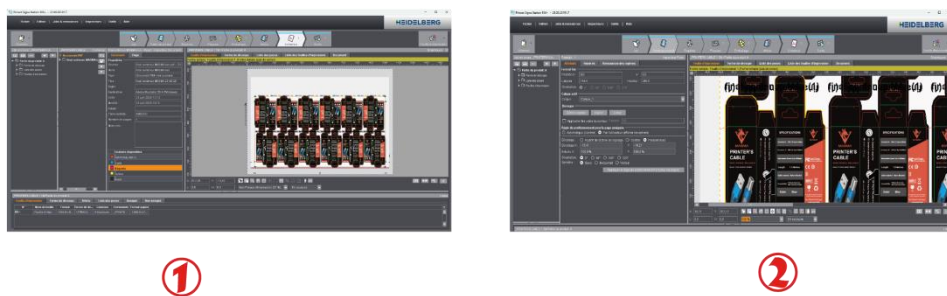


Figure 43. Final verification of the complete printing layout

Source: Taken by the author.

6.8 Saving and Exporting the Final File

The project was saved after all checks were completed. Then the final production file was exported using the required output settings. This file is ready for the next production stage.

This step is important because the exported file will be used for plate making and printing. Saving the project also allows future modifications if needed.



Figure 44. Saving and exporting the final production file

Source: Taken by the author.

7. Printing and Approval of the B.A.T

After finishing the layout, the B.A.T was printed using an HP DesignJet T520 printer. This print was used to check the packaging before production. We checked the text, the images, the colors, and the general position of the design.

The print from this printer is only for checking the work. It is not the final print quality because the final packaging will be printed later by the offset printing machine. The B.A.T helps the operator and the customer find mistakes before production.



Figure 45. HP DesignJet T520 printer used to print the B.A.T for packaging verification

Source: Internet.

8. Plate Making

8.1 Rasterization and File Preparation

After the packaging design was finished, the file was sent to the APOGEE Prepress software. This software prepared the file before plate engraving.

First, the Magma packaging file appeared in the APOGEE interface. The operator checked that the correct file was received before starting the next step.

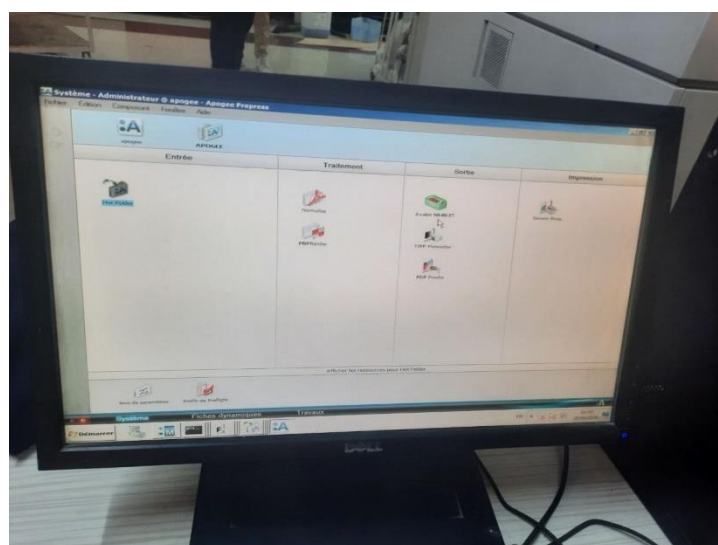


Figure 46. APOGEE Prepress main interface

Source: Taken by the author.

After that, The Magma project was then opened inside the APOGEE job list. The software showed the packaging layout and all the information needed for processing.

This step is important because it confirms that the correct project is ready before plate production.

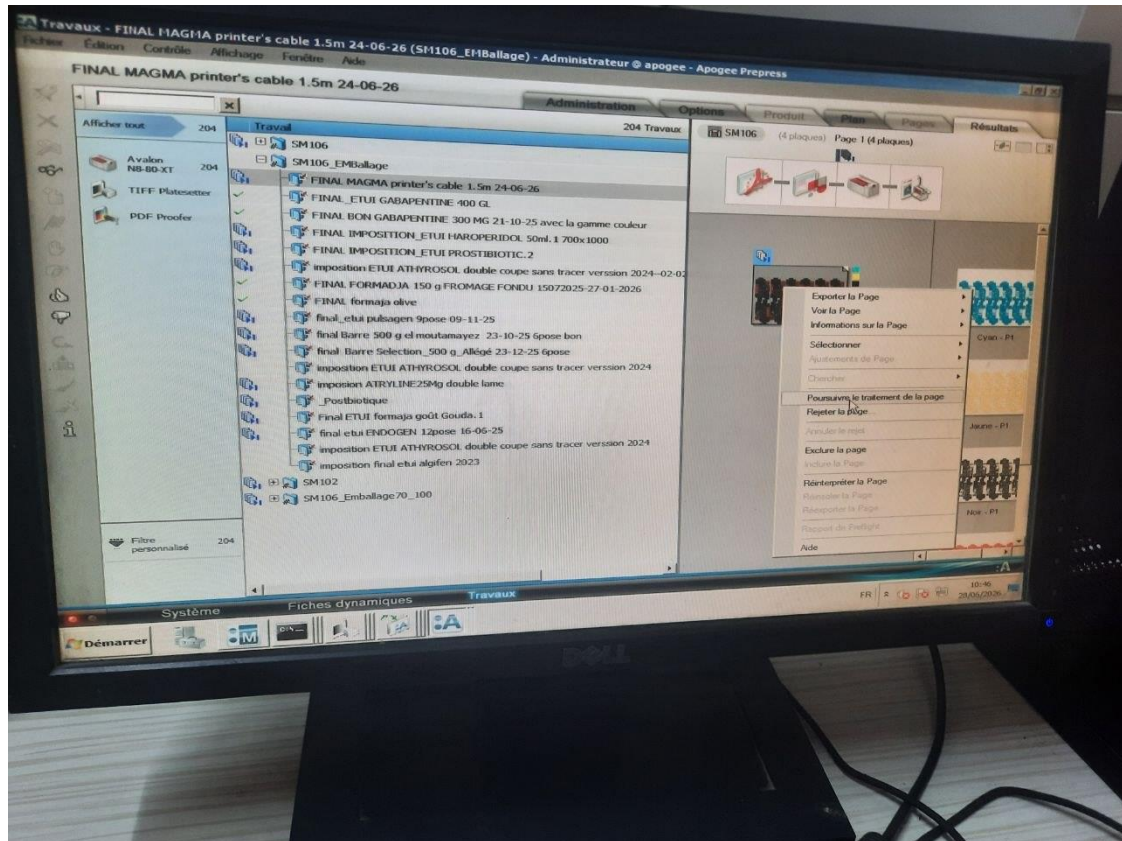


Figure 47. Magma packaging project inside APOGEE

Source: Taken by the author.

After that, the software created the color separation automatically. The design was divided into the four printing colors used in offset printing: Cyan, Yellow, Black, And Pantone 485 C Color.

This step is important because each printing plate carries only one color. The four plates will work together during printing to produce the final image.

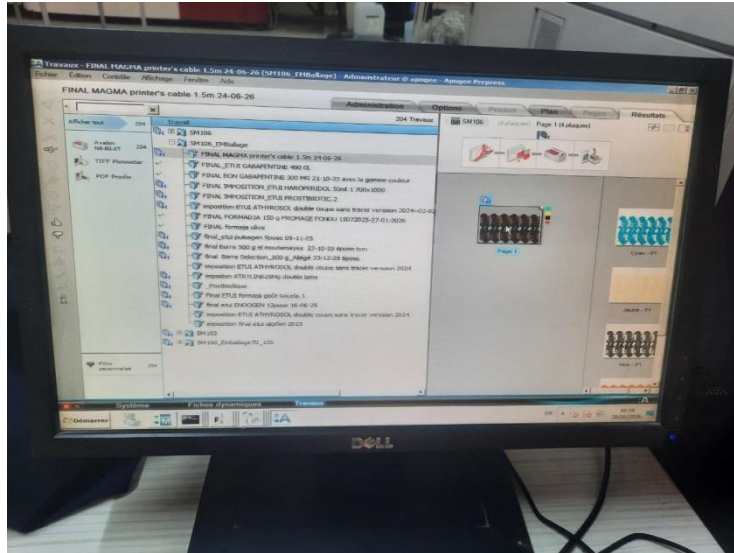


Figure 48. Color separation of the Magma packaging

Source: Taken by the author.

When the separation was checked, the operator selected "Continue Processing" for each plate. This command started the preparation of the engraving process.

This step is important because it sends every separated color to the CTP system for plate production.

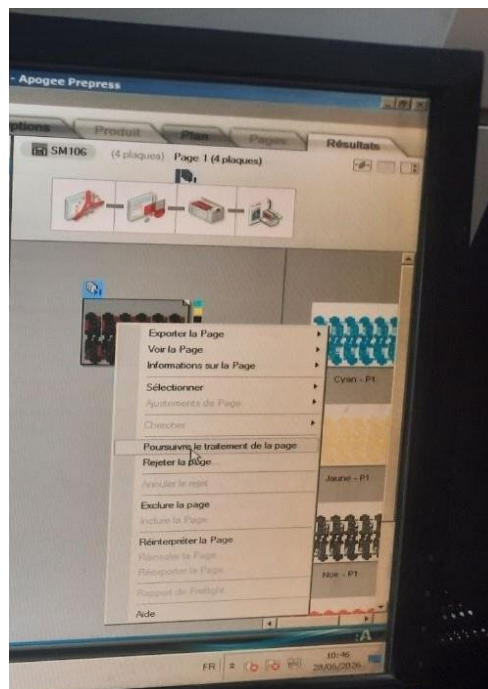


Figure 49. Starting the processing by selecting Continue Processing

Source: Taken by the author.

During processing, the output icon started blinking in green. This showed that the software was sending the plate data to the CTP machine.



Source: Taken by the author.

The CTP machine screen shows all the information needed before plate engraving starts. The operator can see the printing press name, the number of plates to engrave, the number of plates already loaded inside the machine, and the path followed by the plate during the engraving process.

CTP Machine Overview



Figure 51. CTP machine screen showing the plate path inside the machine

Source: Taken by the author.

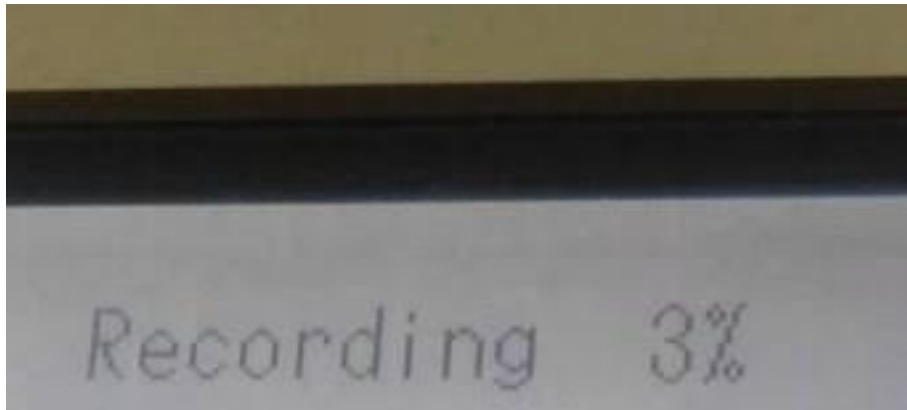


Figure 52. Plate Engraving in Progress

Source: Internet.

For this project, we used the AGFA CTP machine with the Avalon N8-80 processor to produce the printing plates.

The plates used were AGFA Energy Elite Pro with a size of 1055×811 mm, a thickness of 0.30 mm, and a production life of about 500,000 impressions. These positive thermal offset plates are specially designed for thermal CTP systems.

8.3 Plate Engraving

The printing plates were loaded manually into the CTP machine.

For the AGFA Energy Elite Pro plates, the engraving time was about 45 seconds for one plate.

Our packaging uses four printing colors, so the average engraving time for the four plates was about 3 minutes.

This step is important because every printing color needs its own plate before printing can start.

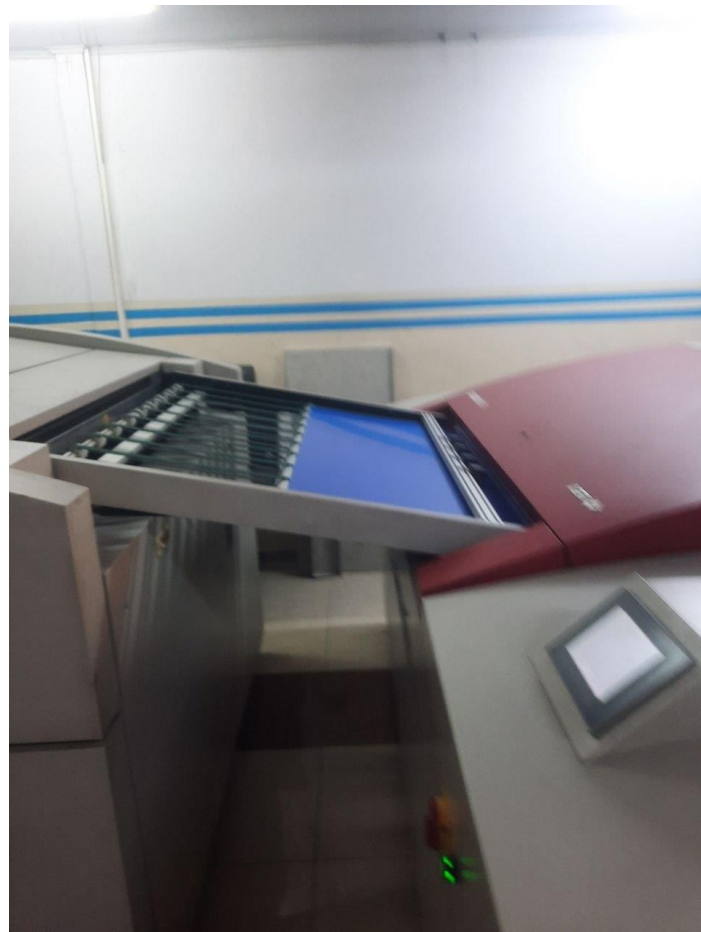


Figure 53. Loading the printing plate into the CTP machine

Source: Taken by the author.

After the engraving process finished, the plate moved automatically to the next section for development.

This transfer is automatic and helps continue the production without stopping the workflow.

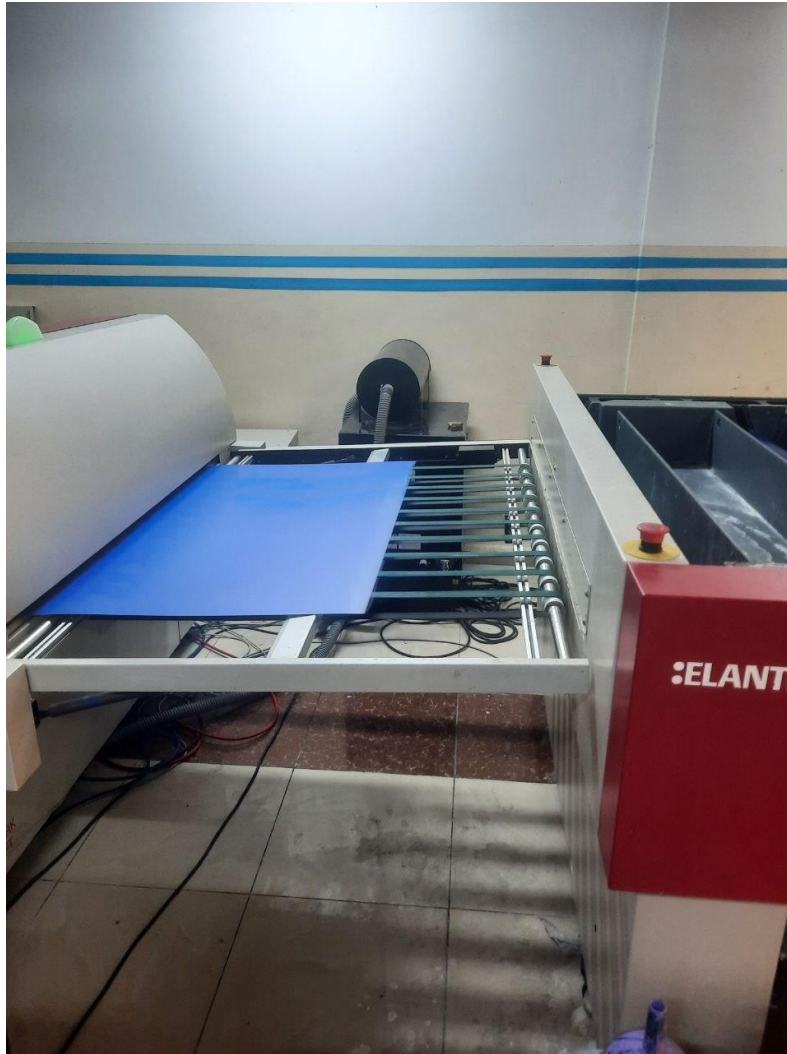


Figure 54. Engraved plate moving to the developing unit

Source: Taken by the author.

8.4 Plate Development

After the plate engraving was finished, the plate entered the developing machine. Before the development process started, the processor used two chemical solutions.

The first solution was the Chemical Developer Solution (Agfa THD200). It develops the printing image on the plate. The second solution was the Plate Finisher and Gumming Solution (Agfa RC795). It protects the plate after development and prepares it for printing.

This step is important because these two solutions help produce a clean and stable printing plate.

CTP Machine Overview



Plate Finisher and
Gumming Solution Agfa RC795



Chemical Developer Solution
(Agfa THD200) used in the CTP processor.

Figure 55. Chemical solutions used in the Avalon N8-80 processor

Source: Internet.

Chemical Solutions Used: Chemical Developer Solution (Agfa THD200) + Plate Finisher and Gumming Solution (Agfa RC795).

The two chemical tanks are connected directly to the Avalon N8-80 processor. During production, the machine uses each solution automatically in the correct stage of the development process.

This helps keep the development process stable and gives the same quality for every printing plate.



Figure 56. Chemical tanks connected to the Avalon N8-80 processor

Source: Taken by the author.

The development temperature was checked on the processor screen before starting the process.

This step is important because the correct temperature helps produce a good printing plate.



Figure 57. Development temperature setting on the Avalon N8-80 processor screen

Source: Taken by the author.

During the development process, the plates were washed, brushed, and dried inside the Avalon N8-80 processor.

These operations prepare the plate for printing and remove all unwanted coating.



Figure 58. Plate development inside the Avalon N8-80 processor

Source: Taken by the author.

After development, the plate came out from the processor and moved to the receiving rack.

The developed plates were stored carefully before being sent to the printing department.



Figure 59. Plate leaving the processor and plate storage

Source: Taken by the author.

8.5 Plate Punching and Plate Bending

After the plates were developed, they were prepared for the printing machine. The same Heidelberg machine was used to punch and bend the plates.

First, holes were made on the plates. These holes help place the plates in the correct position on the printing machine.

After punching, the front edge of each plate was bent. This allows the plate to be fixed correctly on the printing cylinder.

These steps are important because they prepare the plates for correct installation before printing.



Figure 60. Plate punching on the Heidelberg machine.

Source: Taken by the author.

Plate Punching and Plate Bending



Plate Punching



Plate Bending

Figure 61. Plate punching and plate bending

Source: Taken by the author.

8.6 Plate Quality Control

Before printing, the plates were checked using the Techkon SpectroDens device.

The operator selected the screen type, the positive plate type, and the plate manufacturer. Then, the control strip on the plate was measured.

For the Cyan, Yellow, Black colors, the measured values were checked using the standard values.

For the Pantone 485 C color, the Magenta density value was not used because it is a special color. Instead, the Dot Area was checked at 40% and 80%.

This step is important because it helps check the plate quality before printing.

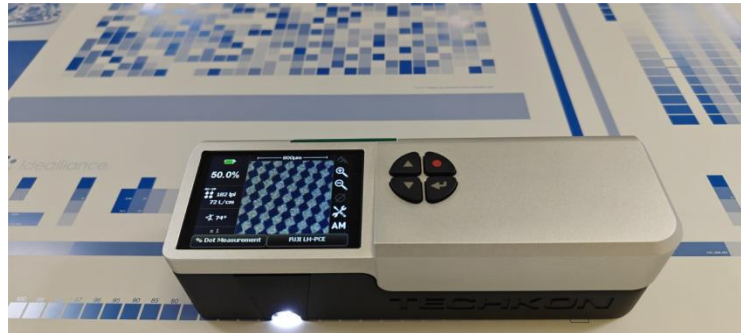


Figure 62. Measurement of the printing plate using the Techkon SpectroDens

Source: Internet.

Chapter Conclusion

In this chapter, we presented the main prepress and plate-making steps used for the Magma Printer's Cable Packaging Box. We prepared the artwork, created the packaging layout, produced the B.A.T, and prepared and checked the printing plates before printing.



SECTION 02 :

PRESS

Chapter Introduction

This chapter presents the printing process of the Magma Printer's Cable Packaging Box. It explains the preparation of the Heidelberg Speedmaster XL 106, the materials used, the make-ready steps, the printing process, and the quality control of the printed sheets.

1. Heidelberg Speedmaster XL 106

We printed our Magma Printer's Cable packaging box on the Heidelberg Speedmaster XL 106 offset printing press.

This is a sheet-fed printing press. It works for packaging printing. It has six printing units and one acrylic varnishing unit.

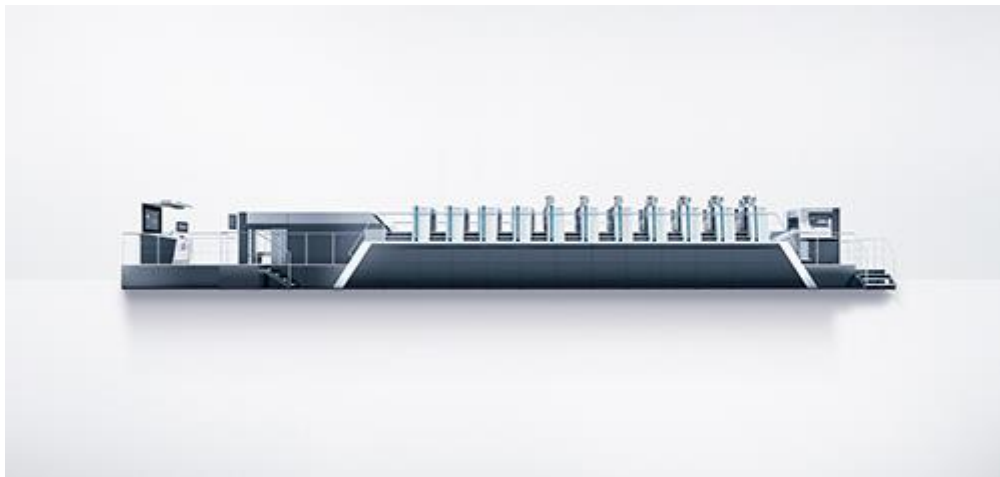


Figure 63. Different units of the Heidelberg Speedmaster XL 106.

Source: Internet.

The different stations of the offset printing press are:

- ✓ Non-stop device
- ✓ Feeder
- ✓ Sheet register
- ✓ Inspection
- ✓ Printing unit
- ✓ AutoPlate
- ✓ Air Transfer system
- ✓ Inking and dampening unit
- ✓ Ink fountain

- ✓ Varnishing unit
- ✓ Dryer
- ✓ Slowdown device
- ✓ StaticStar
- ✓ Delivery system

2. The materials used

For the production of our Magma Printer's Cable packaging box, we used BBTB cardboard.

BBTB cardboard:

The cardboard is made from virgin wood fibres. It is strong and has good mechanical properties. It is good for printing and provides good performance during the production of the packaging.

The cardboard used for our Magma packaging has a grammage of 300 g/m².

3. The printing steps of the packaging

3.1. Job preparation

First, we receive the production file from the prepress department with the engraved plates and the BAT.

The important information in the production file is:

- Paper type: BBTB cardboard
- Grammage: 300 g/m²
- Sheet format: 700 × 1000 mm
- Printing: Recto
- Number of passes: 3
- Number of copies: 20,000 copies
- Varnish type: Acrylic
- Pantone: 485 C

SOCIÉTÉ D'IMPRESSION D'ALGER
DIRECTION COMMERCIALE ET APPROVISIONNEMENTS
ORDRE DE TRAVAIL UNITÉ EMBALLAGE

DOSSIER N° 24/8.226 DATE: 24/06/2026

CLIENT COMMERCIAL: SARL ACT GARY-ET-MICHAEL publicités

DESCRIPTION DU PROJET: Pharmaceutique ☐ AGRICULTURE ☐ AUTRES ☒

CATÉGORIE: Pharmaceutique

TYPE DE PRODUIT: ☒ ETUS ☒ IMPRESSION ☒ découpe ☒ collage ☒

NOTICES: ☐ Impression ☐ découpe ☐ collage ☐

FORMAT DU PRODUIT: 44x33x133 TIRAGE: 20.000

FIN: 44x33x133

NOMINÉ DE COULEURS: 01 COUL. 02 COUL. 03 COUL. 04 COUL. 05 COUL.

COULEUR: 01 COUL. 02 COUL. 03 COUL. 04 COUL. 05 COUL.

PANTONE: 485C

VERNE: ☒ PREFERENCE ☐ NON ☐

VERNE: NON

PELLICULAGE: ☒ OUI ☐ NON ☐

TYPE DE SUPPORT: 8.11.10 Dimensions: 300 400 500

FORMAT DU SUPPORT: 300x400

TYPE DE COULEUR: 300x400

CASSE: ☒ PRELUS ☒ CLASSE ADONDE

CONCOMITANCE: 300x400 300x400 300x400

Carton ou Papier: 300x400 300x400 300x400

Plaquette: 300x400 300x400 300x400

DATE DU BAT: 300x400 DATE DE LIVRAISON PRÉVUE: 300x400

ECHANTILLON: 300x400 300x400 300x400

LAICHE Hafida
Chef d'atelier
Signature

LA COMMERCIALE
Signature

Figure 64. Magma production file

Source: Taken by the author.

3.2. Cleaning of the printing press

Before printing, the offset printing press must be cleaned:

- ✓ Cleaning of the blanket.
- ✓ Cleaning of the inking rollers.

The cleaning of the inking rollers is done in two ways:

Manual cleaning:

It is done using a solvent spray bottle and clean cloths.

Automatic cleaning:

The cleaning is done using a prepared cleaning solution and a special cleaning cloth called a web.



Figure 65. Cleaning of the plate cylinder before printing

Source: Taken by the author.

3.3. Checking the dampening solution

Before starting the printing run, the dampening solution was prepared and checked.

The solution used for the printing press contains:

Water: 86%

Isopropyl alcohol: 10%

Acid additive: 4%

The main parameters were controlled before printing:

Parameter	Standard value	Measured value
Temperature	10–15 °C	10.5 °C
Isopropyl alcohol	10–12%	9%
pH	4.6–5.4	5
Additive	4%	3%
Conductivity	800–1800 $\mu\text{S}/\text{cm}$	1014 $\mu\text{S}/\text{cm}$

We kept the pH at around 5 for offset printing. We must control the pH. If the pH is wrong, it changes the ink-water balance and the print quality.

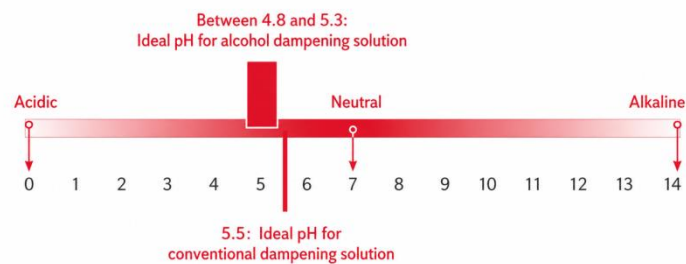


Figure 66. pH scale used to check the acidity of the dampening solution

Source: Designed by the author.



Figure 67. Dampening solutions used for the preparation of the printing press

Source: Taken by the author.

4. Printing Make-Ready

4.1. Resetting the machine

Before starting a new printing job, the control panel is reset.

The control panel is reset, which closes all the ink fountain segments of the five printing units.

The sheet counter is reset.

The registers, feeder, delivery, and the other settings are reset automatically when a new product is introduced into the Heidelberg Speedmaster XL 106.

Note: In general, machine operators do not always reset the machine before every printing job. They may keep the previous register settings depending on the condition of the machine.

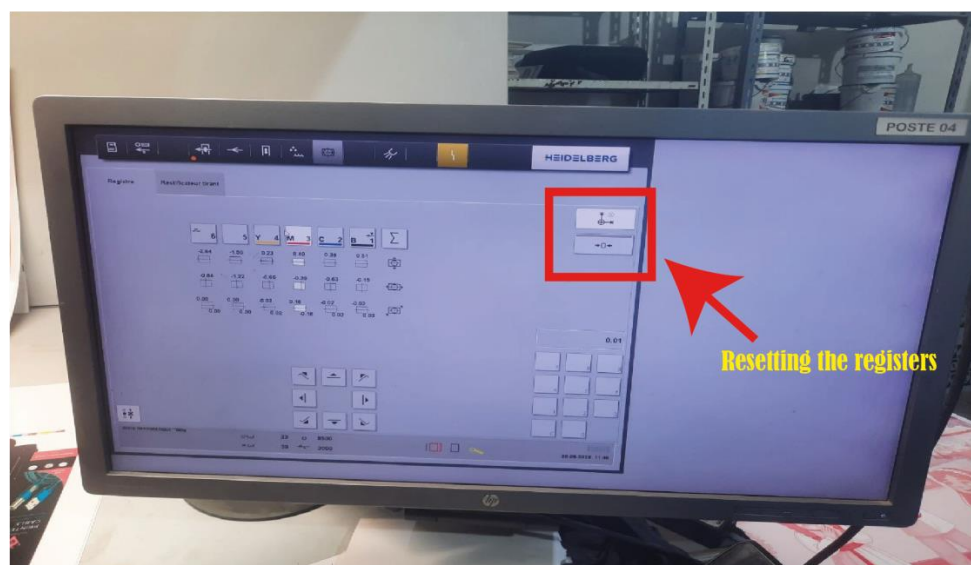


Figure 68. Resetting the registers

Source: Taken by the author.

4.2. Feeder adjustment

The feeder takes the sheets from the pile. It takes one sheet at a time.



Figure 69. Feeder adjustment

Source: Taken by the author.

The air cushions make air under the sheet. This helps the sheet move easily.

PRESSROOM



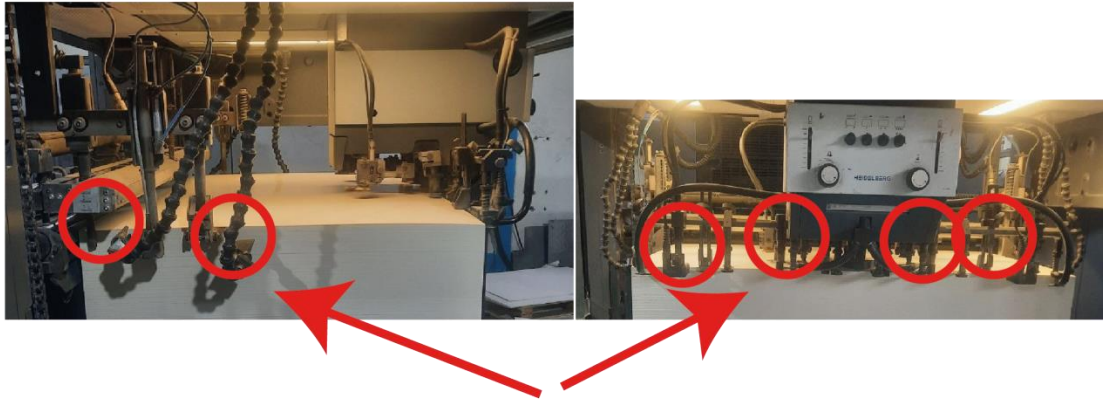
Air Cushions

Figure 70. Air cushion blowers

Source: Taken by the author.

The sheet separation blowers separate the sheets. They lift 5 to 10 sheets.

PRESSROOM



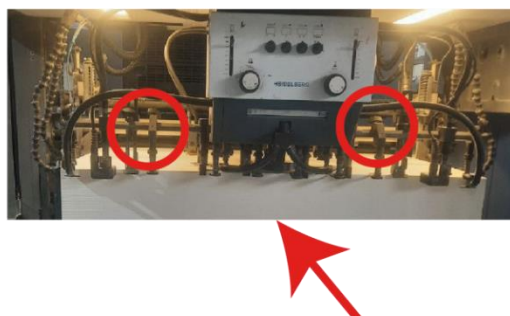
Sheet separation blowers

Figure 71. Sheet separation blowers

Source: Taken by the author.

The separation blades stay in the sheet pile. Setting: 4 to 10 mm. They help stop two sheets together.

PRESSROOM



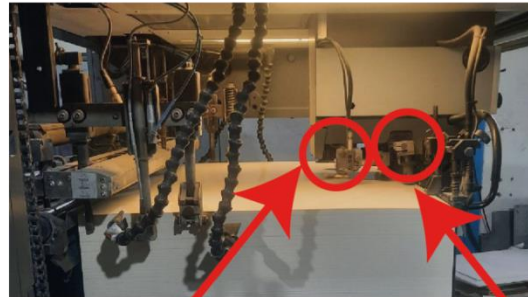
Separation blades

Figure 72. Separation blades

Source: Taken by the author.

The lifting suckers lift the sheet from the pile. They touch the sheet during lifting. The transport suckers take the sheet from them. Height: 5 mm.

PRESSROOM



Lifting suckers

Forwarding suckers

Figure 73. Lifting suckers and Transport suckers

Source: Taken by the author.

The presser feet keep the sheet in the right position. The mark stays aligned with the end of the sheet pile.

PRESSROOM



Presser feet

Figure 74. Presser feet

Source: Taken by the author.

Double-sheet detector

The machine has two double-sheet detectors.

They detect two sheets at the same time.

The machine stops when two sheets come together.

The detector changes automatically for the sheet grammage.

Side guides

The side guides keep the sheets in the correct position during printing.

4.3. Delivery adjustment

We set the delivery for the printing format.

We make the printing format smaller by 3 mm, and we make the rear guides smaller by 5 mm.

This keeps the sheet pile in the right place.

The sheet slow-down devices use suction to slow the sheet down and pull it flat.

This helps the sheet drop into the pile easily.

We increment the fan airflow to help the sheet drop onto the pile and stop it from floating.

The blow tubes send air to some spots on the sheet.



Figure 75. Delivery

Source: Taken by the author.

4.4. Sheet registration (blank run)

Every sheet must stay in the exact place. It must stay exactly in line from front to back, and exactly from side to side. This helps the sheet move easily on the register table and go into the printing machine without problems.

The transport table has air holes. The air pulls the paper down. This keeps the paper in its place.

The sheets move one after another. Each sheet rests a little on the end of the sheet before it.

We set the side guide to the sheet width. With the front lays, this makes sure every sheet is in the same place.

Good sheet registration saves time. It stops us from losing time during the printing run.

We do a blank run (no ink). This is to check the sheet movement from the start to the end of the machine.

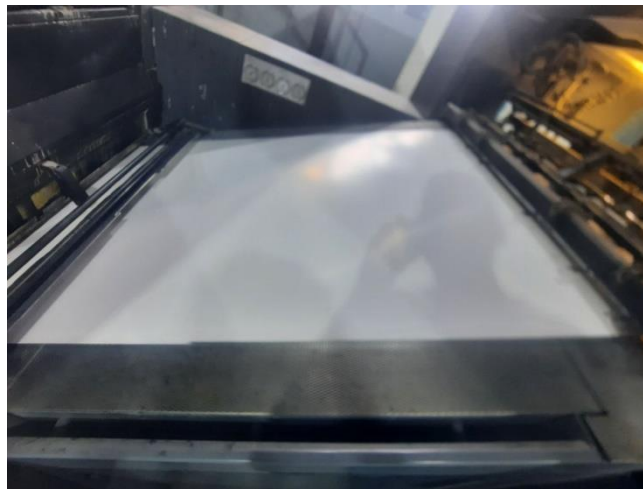


Figure 76. Blank run

Source: Taken by the author.

4.5. Production file

The production job is loaded from the available data on the control panel.

The sheet format can also be entered or changed in the Printing Functions menu. The sheet direction and paper thickness can also be set there.

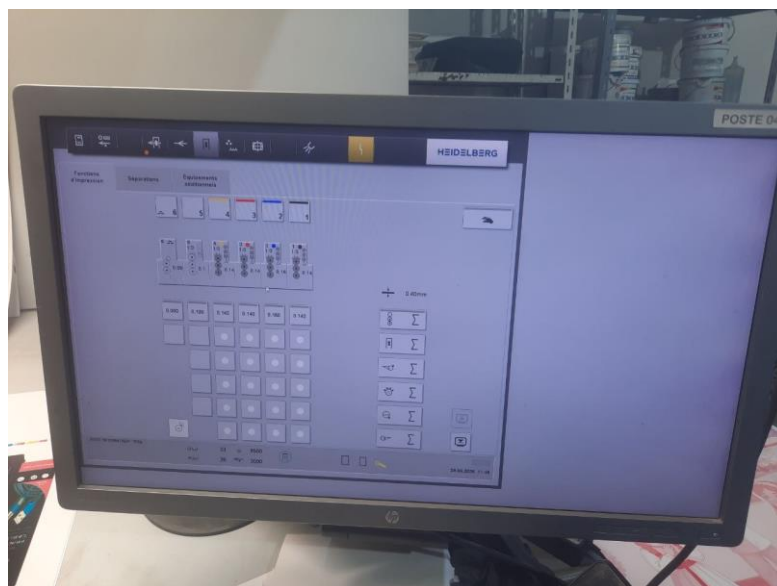


Figure 77. Printing Functions settings

Source: Taken by the author.

4.6. Work data and printing speed

The work data, print run and printing speed are set on the control panel. For Magma, the printing speed is set at 8500 sheets/hour.

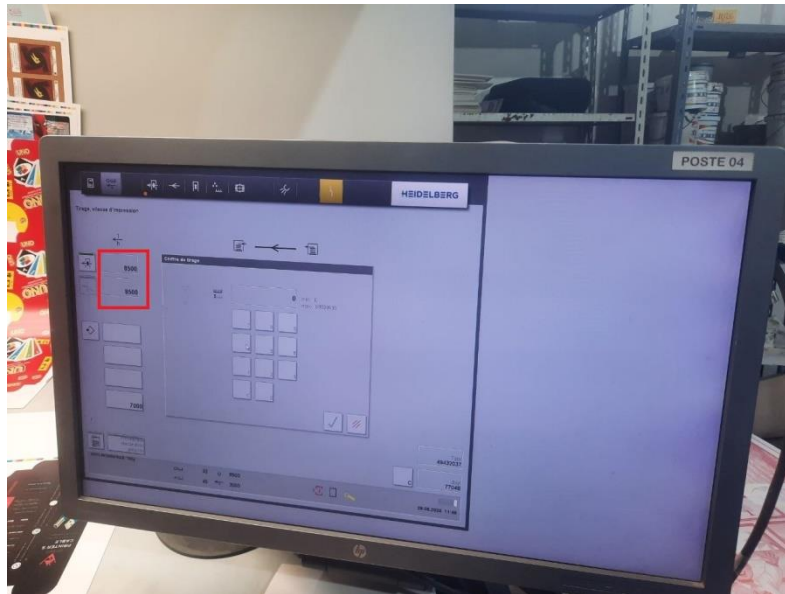


Figure 78. Work data and printing speed

Source: Taken by the author.

4.7. Printing data adjustment

The sheet size is set to 1000 × 700 mm.

The paper thickness is set to 0.4 mm.

The left or right position of the paper is set.

The paper turning system and the transfer part are also set.

Then, the sheet feeding is adjusted so the paper moves correctly in the machine.

4.8. Inking data adjustment

For manual ink profile reading, a printing plate is placed on the light table.

The printable and non-printable areas are checked. This helps set the ink amount for each printing unit.

The ink profile tab is then opened on the control panel. The ink flow is adjusted with the ink fountain keys and the ductor speed, based on the ink areas of the plate.

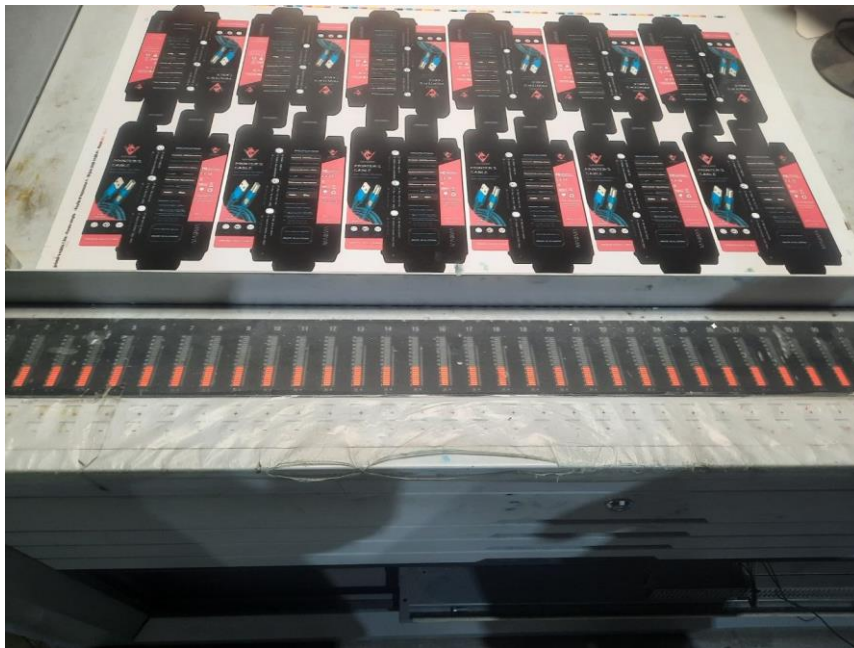


Figure 79. Manual ink profile reading on the light table.

Source: Taken by the author.

The ink profile is then set on the control panel for each printing unit.



Figure 80. Ink profile tab

Source: Taken by the author.

5. Printing Part:

5.1. Powder and dryer adjustment

The powder and dryer are turned on before printing using the control panel.

The anti-set-off powder helps prevent the ink from marking the other sheets.



Figure 81. Anti-set-off powder

Source: Internet.

5.2. Ink fountain filling

The ink fountains are filled with the correct ink for each printing unit.

For Magma, the inks used are Cyan, Yellow, Black and Pantone 485 C.



Figure 82. Ink fountain filling

Source: Taken by the author.

5.3. Plate cleaning

The plates are cleaned with a cleaner.

This removes dust and stains from the plates.

The cleaning is done before placing the plates on the plate cylinder. It helps the ink stay on the printable areas.



Figure 83. Cleaner

Source: Internet.

5.4. Plate mounting

The plate is placed on the machine cover.

The printing unit is selected.

The plate cylinder is put in position.

The plate is placed on the pins of the plate cylinder.

The clamp is closed. The cylinder is turned to check the plate position.

The printing unit is deselected. The cover is then closed.



Figure 84. Plate mounting on the offset press

Source: Taken by the author.



Figure 85. Printing unit control buttons.

Source: Taken by the author.

5.5. Checking the first printed sheets

The first printed sheets are checked on the control panel.

The color registration is checked. The colors must be in the correct position.

A loupe is used to check the cutting marks. If there is a problem, the color position is adjusted for each color.

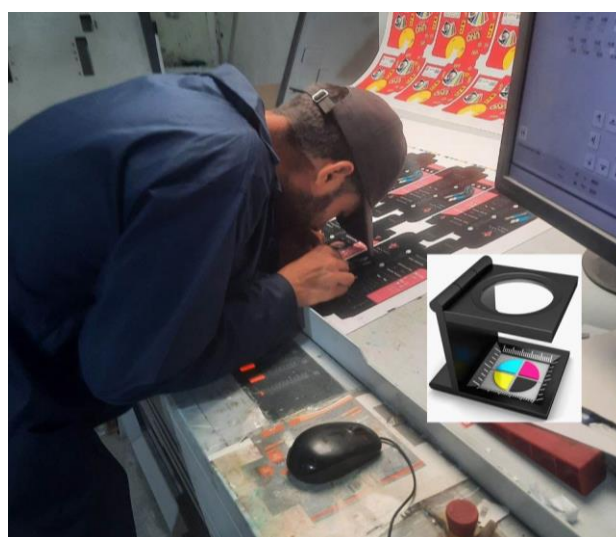


Figure 86. Loupe

Source: Taken by the author.

The color percentage on the color control strip is also checked.



Figure 87. Color registration

Source: Taken by the author.

The registers are checked during the printing run. This helps keep the colors in the correct position.

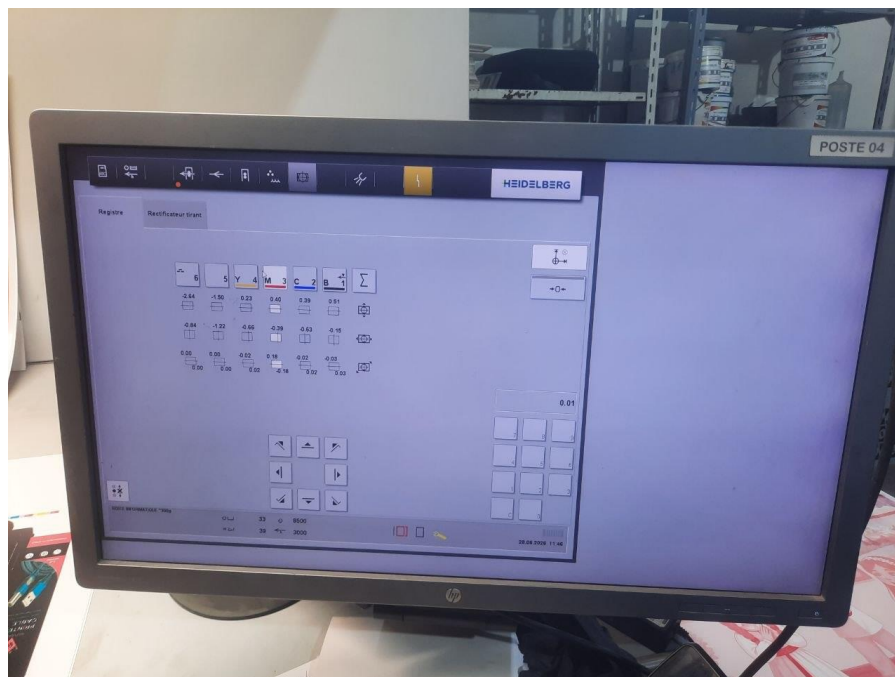


Figure 88. Register menu

Source: Taken by the author.

5.6. Inking and dampening

The Inking and Dampening menu controls the ink and dampening solution.

The ink flow is set for each printing unit on the control panel. The ink setting is between 20% and 45%. The machine changes the ink film width, making it more or less.

The dampening solution level and air speed are also adjusted on the control panel. The setting is between 20% and 40% for each printing unit.

The quantity of ink on the paper is checked during printing. It is checked by eye and can be changed from the control panel. This changes the ink roller speed and the ink controls.

The goal is to have a good quantity of ink on the paper and less ink transfer and drying problems.

5.7. Pre-inking

The pre-inking menu is used before printing.

It prepares the ink unit after cleaning.

The menu allows the operator to:

- ✓ close the ink zones at the end of the job.
- ✓ start the pre-inking process.
- ✓ select the printing units.
- ✓ set the number of ink movements.
- ✓ control the pre-inking of the selected printing units.

The number of ductor strokes controls the ink amount during pre-inking.

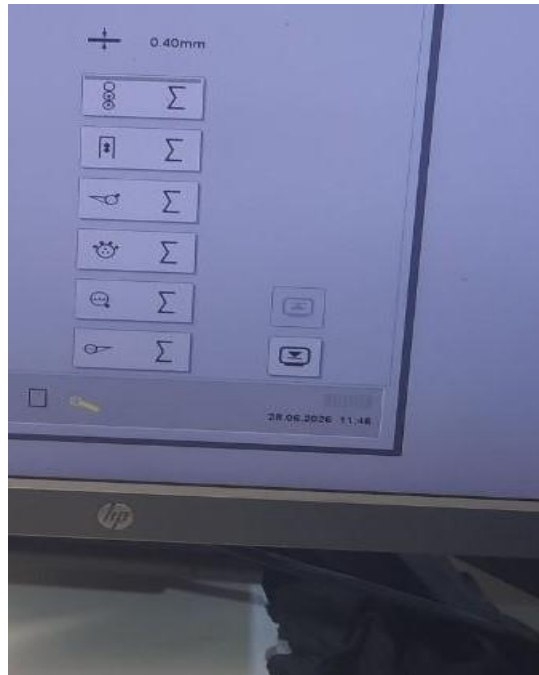


Figure 89. Pre-inking settings

Source: Taken by the author.

5.8. Dampening system

The dampening system is built into the printing machine. It needs cooling and alcohol control.

Before printing, the dampening solution is checked on the control panel.

The values are:

- ✓ Water: 86%
- ✓ pH: 5
- ✓ Temperature: 10.5 °C
- ✓ Alcohol: 9%
- ✓ Additive: 3%

The dampening system is turned on with a button on the machine. Its settings can also be checked and changed on the control panel screen.

The dampening solution is acidic, with a pH of about 5.

When the pH is below 4.8:

- ✓ the ink dries faster.
- ✓ the machine can get rusty.
- ✓ ink and water mix too much.
- ✓ paper can have problems.

When the pH is above 5.5:

- ✓ the plate can lose its good surface.
- ✓ ink and water mix too much.

When the temperature is too high:

- ✓ the alcohol evaporates faster.
- ✓ the solution is less thick.
- ✓ ink and water mix more.

When the temperature is too low:

- ✓ You can see water under the tanks.
- ✓ the solution is more thick.
- ✓ ink and water mix more on the plate.

5.9. Varnishing preparation

- Three packing sheets are used. Each one is 0.4 mm.
- The total thickness is 1.20 mm.
- One more packing sheet of 0.1 mm is added.
- A varnish blanket of 1.95 mm is used.
- The packing sheets are cut 5 mm smaller on each side.
- The sheets are placed under the varnish blanket.
- We place the blanket on the blanket cylinder and use a torque wrench to make sure it does not move.

PRESSROOM



Figure 90. Varnishing preparation

Source: Taken by the author.

5.10. Drying and anti-set-off system

The drying unit is activated from the control panel.

The anti-set-off powder system is also activated.

The powder helps prevent ink contact on the back of the next sheet.

This helps keep the back of the sheet clean.

5.11. Checking the print

The printed sheet is checked for print quality and color registration.

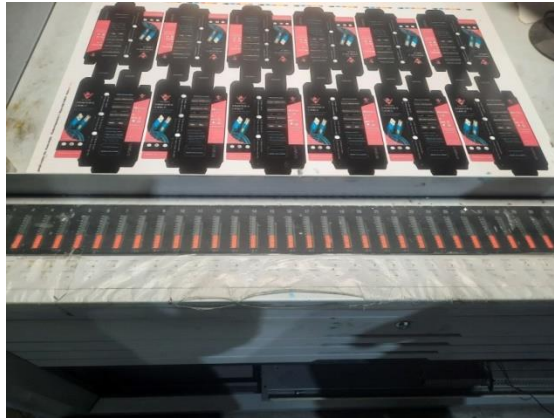


Figure 91. Printed sheet

Source: Taken by the author.

The X-Rite densitometer checks the ink density on the color control strip.

First, the device is set on the white paper.

Then, it is set on the color control strip.

The values are compared with the standard values.

The dot gain is also checked.

The printed sheets are checked again and again during the print run.

This helps control the ink and registration.



Figure 92. Ink density measurement with X-Rite densitometer

Source: Taken by the author.

5.12. Problems during printing

Problem	Cause	Solution
Picking	Ink drying on the blanket.	Clean the blanket with a cloth and blanket cleaner 7400
Set-off	More ink or slow drying	Turn on the anti-set-off system and increase drying
Ghosting	Blanket not clean	Clean the blanket with a cloth and blanket cleaner 7400
Not enough water	Low water flow	More water flow
Varnish stains	The varnish is hard.	Add a product to make the varnish easier to use.
Paper waviness	Poor paper storage	Keep the paper in a less humid place

Table 3. Problems, causes and solutions during printing

Chapter Conclusion

In this chapter, we presented the main printing steps of the Magma Printer's Cable Packaging Box. We prepared the printing press, adjusted the materials and settings, printed the packaging, and checked the print quality and common printing problems.



SECTION 03 :

POST-PRESS

Introduction

This chapter presents the finishing stages of the Magma Printer's Cable Packaging Box. It explains the die-cutting process, waste separation, folding and gluing, and the final packaging of the finished boxes.

1. Die-cutting

1.1. Die-cutting machine



Figure 93. MK Easymatrix die-cutting machine

Source: Taken by the author.

1.2. Technical data

Parameter	Value
Minimum sheet size	400 × 350 mm
Maximum cutting format	1040 × 730 mm
Maximum sheet size	1050 × 740 mm
Production speed	7500 sheets/hour
Feeder pile height	1650 mm
Power	75 kW
Air pressure	6 bar
Air compressor capacity	15 m ³ /h
Air tank capacity	100 L
Machine size (L × W × H)	6.13 × 4.28 × 2.45 m
Machine weight	16 tonnes
Minimum gripper margin	9.5 mm
Die-cutting force	300 tonnes

1.3. Die-cutting form

The die-cutting form is made on a wooden board.

The board size is 1040 × 725 mm.

The board thickness is 18 mm.

The form has different parts:

- Channels: help the cardboard fold.
- Foams: help remove the waste from the cutting blades.
- Cutting blades: cut the cardboard in the required shape. Their thickness is 23.8 mm.
- Creasing rules: make the fold lines. Their thickness is the same as the cutting blades.
- Braille matrix and Braille counterpart (dots you can feel): make the Braille.

For the Magma box, the cutting and creasing parts are placed inside the form for the box design.



Figure 94. Die-cutting form

Source: Taken by the author.

1.4. Calculation of the creasing counterparts

The creasing system is calculated using the cardboard thickness and the creasing rule.

For our cardboard, the material thickness is 0.50 mm.

The creasing channel and the creasing rule are calculated to get the correct folding.

The optimum creasing system is:

$$0.50 \times 1.50 \text{ mm}$$

The optimum creasing rule is:

$$2 \text{ pt} \times 23.30 \text{ mm}$$

POST-PRESSE

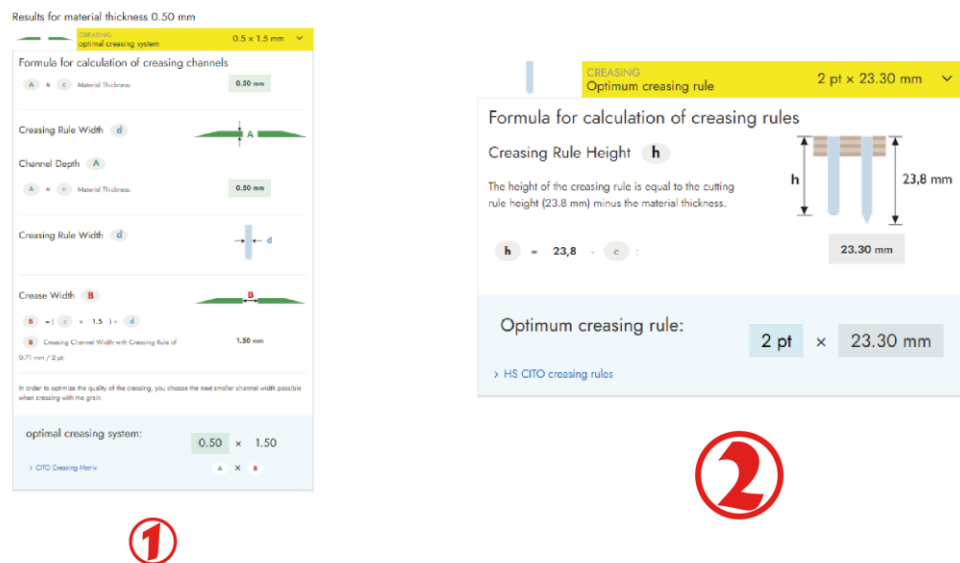


Figure 95. Creasing calculation and optimum creasing system

Source: Personal photograph.

1.5. Materials used

The guillotine cuts the creasing counterparts to the required size.

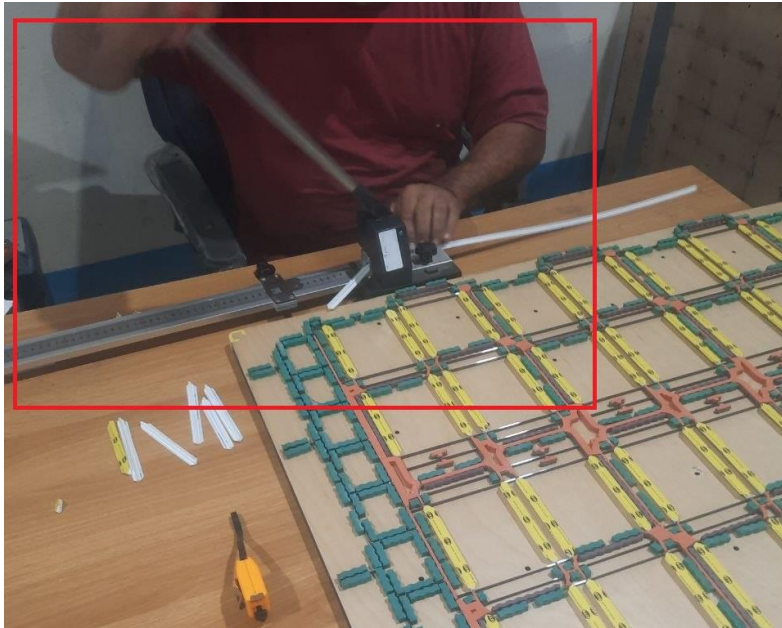


Figure 96. Guillotine

Source: Taken by the author.

The creasing counterparts help make clean crease lines. They also help the cardboard fold easily.

POST-PRESSE

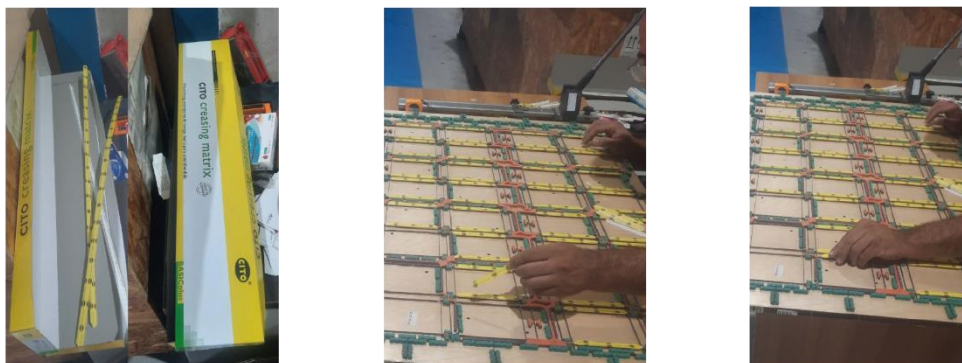


Figure 97. Creasing counterparts

Source: Taken by the author.

The foams help remove paper pieces from the cutting blades.

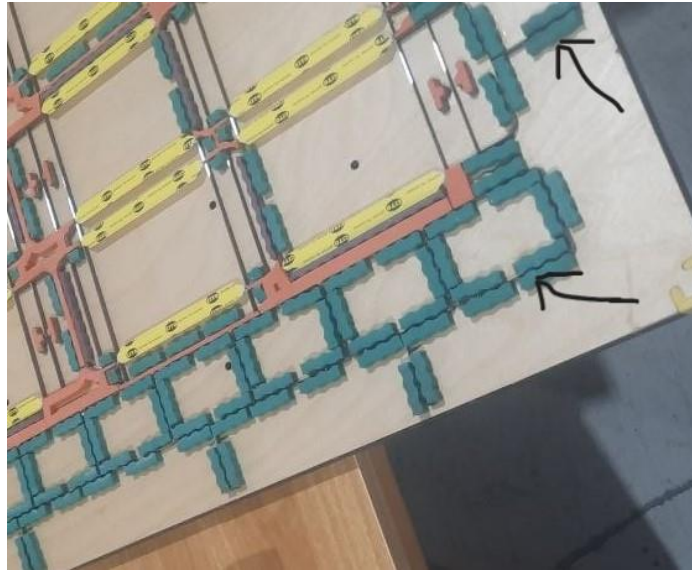


Figure 98. Foams

Source: Taken by the author.

The adhesive tape adds pressing to the creasing counterpart. It also helps correct small problems.



Figure 99. Adhesive tape

Source: Internet.

1.6. Die-cutting form preparation

The machine is cleaned before the die-cutting form is put in place.

The platen is taken out and cleaned. This helps get good cutting and creasing.

The creasing counterparts (channels) are placed on the creasing rules.

The adhesive protection is removed after placing the counterparts on the frame.



Figure 100. Die-cutting form with channels

Source: Taken by the author.

The die-cutting form is fixed on the platen frame.

The makeready sheet is placed on the platen and aligned with the center line.

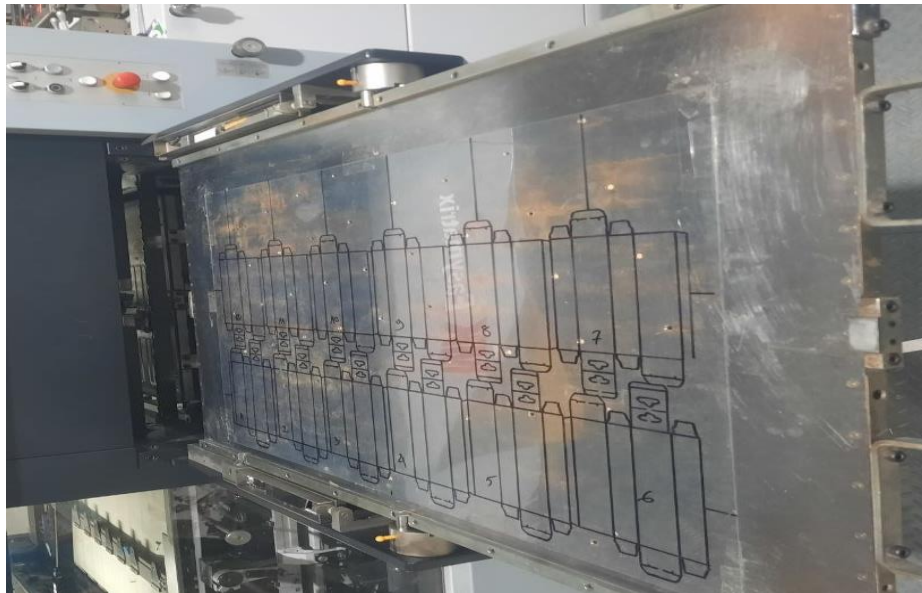


Figure 101. Makeready sheet

Source: Taken by the author.

The frame is pressed. The creasing counterparts stick to the lower part.

The frame is taken out. The protections are removed.



Figure 102. Creasing rule protection

Source: Taken by the author.



Figure 103. Braille counterparts and channels

Source: Taken by the author.

1.7. Machine settings

The feeder is adjusted for the sheet size 700 × 1000 mm.

The front lays and side guides are adjusted to keep the sheet in the correct position.

The register table is also adjusted.

The die-cutting force is set at 0.42 Nm.

The machine speed is set at 3317.

The cutting force setting is important for good cutting, machine production and die-cutting form life.



Figure 104. MK Easymatrix 106 CS control panel – machine settings.

Source: Taken by the author.

1.8. Feeder adjustment

The feeder pallet is placed in the center.

The feeder head is set for the paper size.

The presser foot is set at the end of the paper.

The air blowers are adjusted.

The lifting suckers are adjusted.

The transport suckers are adjusted.



Figure 105. Feeder adjustment

Source: Taken by the author.

POST-PRESSE

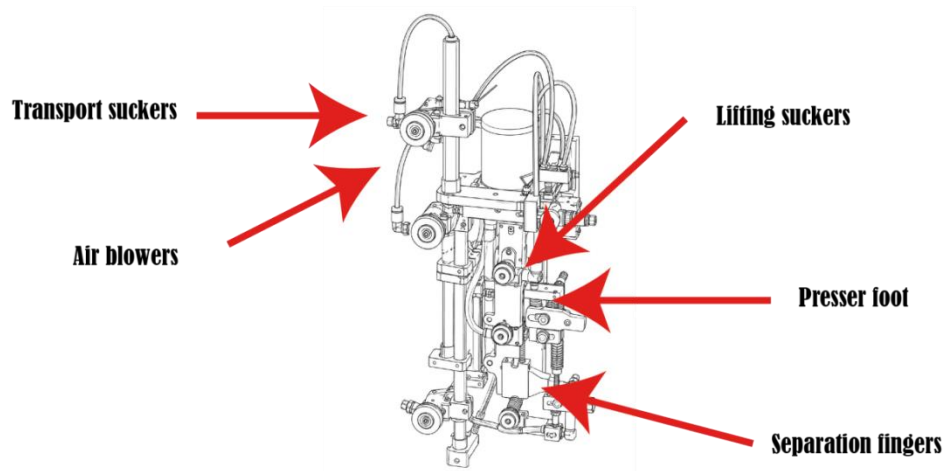


Figure 106. Feeder head

Source: Taken by the author.

1.9. Front and rear rollers adjustment

The front and rear rollers are adjusted.

They help stop the sheets from moving back.



Figure 107. Feed table roller adjustment.

Source: Taken by the author.

The delivery rear stop is adjusted for the sheet.



Figure 108. Delivery rear stop adjustment

Source: Taken by the author.

1.10. Waste separation

After die cutting, the paper stack is moved to the waste separation stage.

This stage removes the waste parts from the die-cut sheets.

The waste is removed by hand or with a waste separation machine.

Adhesive makeready strips are also used in weak cutting areas.

This helps improve cutting and keep the cutting rules in good condition.

1.11. Manual waste separation

After die cutting, the product goes through manual waste separation.

The cutting links are removed by hand.

The sheets are separated from the waste parts.

The waste is removed, and the products are prepared for the next stage: folding and gluing.



Figure 109. Waste separation machine – 4500 rpm.

Source: Taken by the author.

POST-PRESSE



Figure 110. Manual waste separation

Source: Taken by the author.

1.12. Problems during die cutting

Problem	Cause	Solution
Sheet skew	Weak cutting links, bad foam setting, bad placement	Add cutting links; adjust the foams, adjust the placement
Creasing counterpart break	Wrong channel size, bad adhesive	Check the channel size, check the adhesive and change it if needed
Uneven cutting	Low cutting force	More cutting force, check the makeready.

Table 4. Problems, causes and solutions during die cutting.

Folding and gluing

2. Folding and gluing machine

After waste separation, the box goes to the folding and gluing machine.

Before starting, the machine is adjusted from the feeder to the delivery.



Figure 111. MK Diana Easy folding and gluing machine

Source: Taken by the author.

2.1. Technical data

Parameter	Value
Material range	200–1200 g/m ²
Flute types	N, E, F and BC
Speed	300 m/min
Glue type	Water-based
Feeder width	340–1100 mm

Machine size (L × W × H)	17400 × 1850 × 1500 mm
Total weight	9600 kg
Total power	26 kW
Air compressor capacity	10 m ³ /h



Figure 112. MK Diana Easy

Source: Taken by the author.

2.2. Glue used

For the Magma box, we used COLLE RE: MR 432 glue.



Figure 113. Glue used for the Magma box.

Source: Taken by the author.

2.3. Initial feeder adjustment

- ✓ The first sheet is placed on the machine center line.
- ✓ The side gauge is set to the center of the sheet.
- ✓ The main gauge is set on a transport belt and in line with a lower roller.
- ✓ The auxiliary gauge gives extra support.
- ✓ The transport belts are set for the sheet size.
- ✓ The rear supports are set with a small angle for good contact with the sheet.
- ✓ The transport suckers are also adjusted for good sheet movement.
- ✓ The sucker pressure is set by hand.
- ✓ The other supports are set with a 10–20 mm distance from the transport belt for the high cardboard weight.
- ✓ The carriages are adjusted.
- ✓ The side cheeks are set at 2 mm.

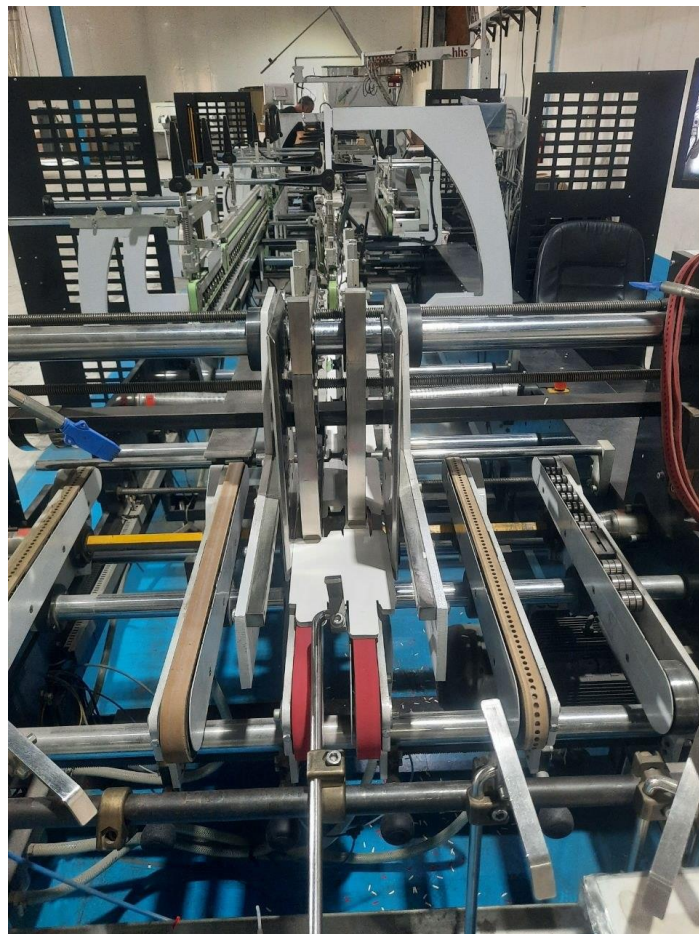


Figure 114. First sheet placement and feeder adjustment

Source: Taken by the author.

2.4. Pre-breaking of folds

The pre-breaking helps make the folding easier.

The Magma box has 2 folds.

The first pre-breaking is for the glue flap.

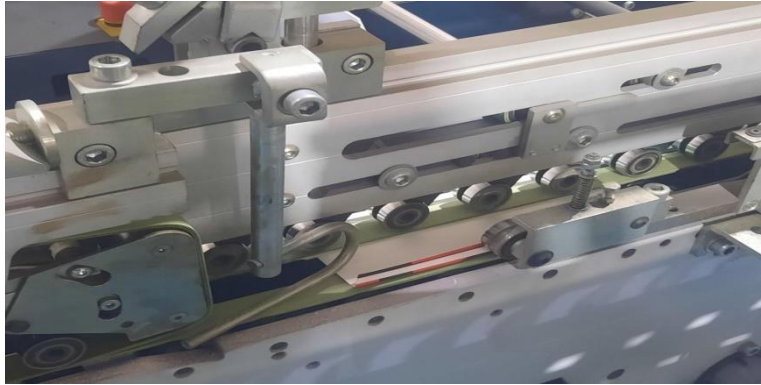


Figure 115. First pre-breaking

Source: Taken by the author.

The second pre-breaking is for the second fold.

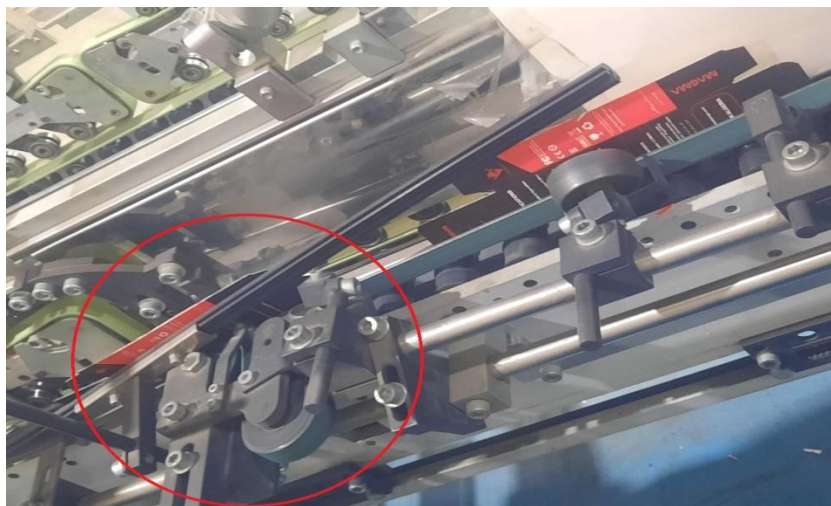


Figure 116. Second pre-breaking.

Source: Taken by the author.

After each pre-breaking, the box opens again.

The transporters keep the box in position during the process.



Figure 117. Box passage through the transporters

Source: Taken by the author.

2.5. Gluing adjustment

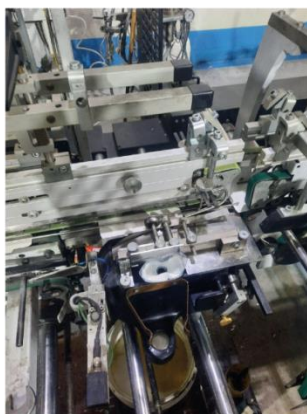
The gluing system has an upper glue tank, a lower glue disc and a scraper.

The glue trace, glue stability and gluing distance are adjusted.

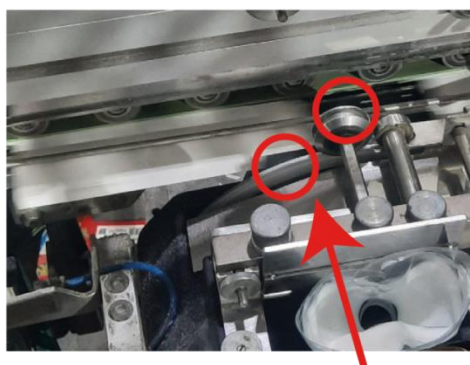
The die-cut sheet goes through the gluing unit.

It passes between the glue disc and the upper glue part.

Gluing system



Glue tank



upper and lower glue discs

Figure 118. Gluing system

Source: Taken by the author.

2.6. Folding adjustment

The folders are on the left and right side.

They fold and close the boxes.

The folders use upper and lower transporters.

The upper transporters move the box to the 90° position.

The box moves by contact between the folding belts and the sheet.

After folding, the box goes to the press.

The press holds the box and helps the glue stick.

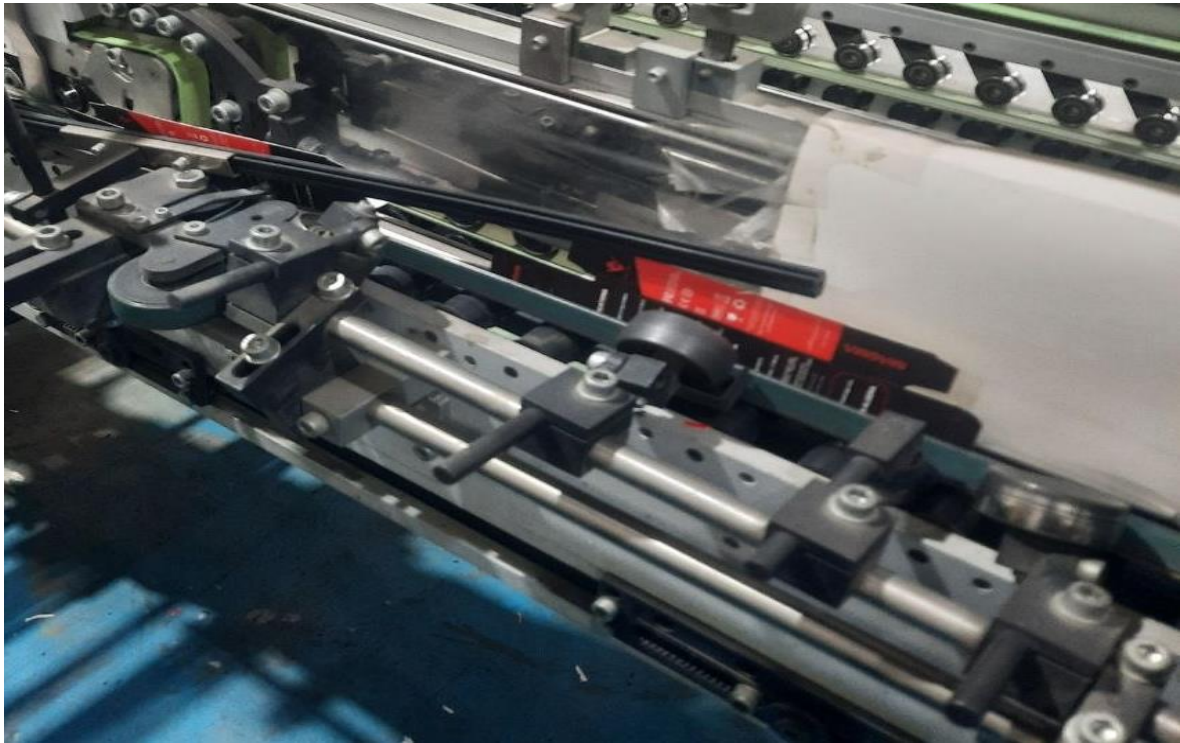


Figure 119. Box closing (folding)

Source: Taken by the author.

Folding adjustment

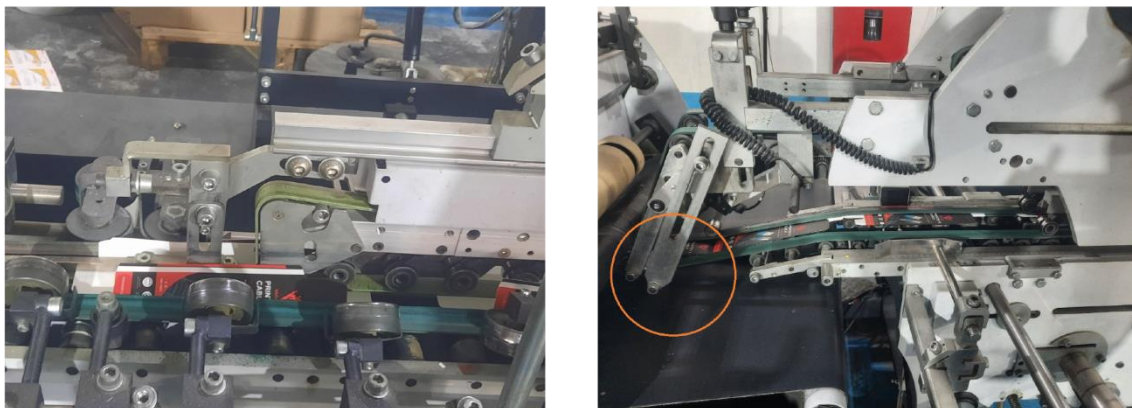


Figure 120. Box passage through the transporters

Source: Taken by the author.



Figure 121. Box passage through the press.

Source: Taken by the author.

2.7. Press settings

The following settings are adjusted from the control panel:

Separation hammer: counts the number of boxes in each box.

Distance between boxes: sets the space between boxes.

Press belt speed: the speed is changed for the box size.

Box model: the model is selected by the number of glue points.

For the Magma box, the selected model has 1 glue point.



Figure 122. Folding and gluing machine screen settings.

Source: Taken by the author.

2.8. Packaging and delivery



Figure 123. Filling the cartons with the finished boxes

Source: Taken by the author.

The cartons are then prepared for delivery and placed on pallets.



Figure 124. Cartons ready for delivery.

Source: Taken by the author.

The finished Magma box is then ready for delivery.



Figure 125. Final Magma product

Source: Taken by the author.

2.9. Problems during folding and gluing

Problem	Cause	Solution
Bad sheet entry	Rear supports press the sheet side	Adjust the supports
Bad folding	Folders	Adjust the folders
Too much glue	Scraper opening	Adjust the glue trace

Table 5. Folding and gluing problems

Chapter Conclusion

In this chapter, we presented the main finishing steps of the Magma Printer's Cable Packaging Box. We prepared the die-cutting form, cut and separated the cardboard, folded and glued the boxes, and prepared the finished products for delivery.

General Conclusion

GENERAL CONCLUSION

This work helped us understand the main steps of the printing and packaging process.

We studied the different stages, from pre-press to printing and post-press.

The practical work helped us understand how the machine settings affect the quality of the final product.

We also learned how to check ink, registration, cutting, creasing, folding and gluing.

This practical experience helped us improve our technical skills and better understand the work in the printing industry.

The work also showed us the importance of correct machine settings, regular checking and good product quality.

This experience is a good preparation for our future professional work.

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Online Search Tools

- Google. *Google Search*. [Google Search](#)
- Yandex. *Yandex Search*. [Yandex Search](#)

Artificial Intelligence Tools

- OpenAI. *ChatGPT*. [ChatGPT](#)
- Google. *Gemini*. [Google Gemini](#)
- DeepSeek. *DeepSeek*. [DeepSeek](#)



ANNEXES

Annex No. 1: Manufacturing File

[illegible]

PROTOCOLE DE CONTRÔLE DU PRODUIT

PREPRESSE :

- Consulter le dossier technique/commercial
- Contrôle de l'imposition simple ou en amalgame
- Contrôle du tracé
- Signature du bon à graver (nom, prénom du signataire)

NOM ET PRENOM:

DATE:

SIGNATURE:

IMPRESSION :

- Consulter le dossier technique/commercial
- Contrôle du bon à tirer préresse
- Contrôle de la conformité de la matière par rapport au dossier technique /commercial
- Contrôler l'imposition
- Contrôler l'encrage (conformité, repérage, vernis s'il ya lieu....)
- Signature du bon à imprimer (nom, prénom du signataire)

NOM ET PRENOM:

DATE:

SIGNATURE:

DECOUPE :

- Consulter le dossier technique/commercial
- Contrôle du bon à tirer impression
- Contrôle de la forme de découpe et du tracé
- Contrôle du sens et de la qualité de la découpe
- Signature du bon à découper (nom, prénom du signataire)

NOM ET PRENOM:

DATE:

SIGNATURE:

Collage et pliage :

- Consulter le dossier technique/commercial
- Contrôle du bon à découper
- Contrôle du pliage
- Contrôle du collage (qualité, nombre de points, ouverture...)
- Signature du bon à relier (nom, prénom du signataire)

NOM ET PRENOM:

DATE:

SIGNATURE:

Annex No. 2: Previous Prints Cabinet

